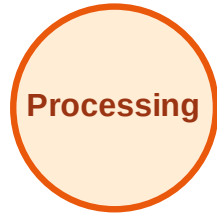


BFS Traversal

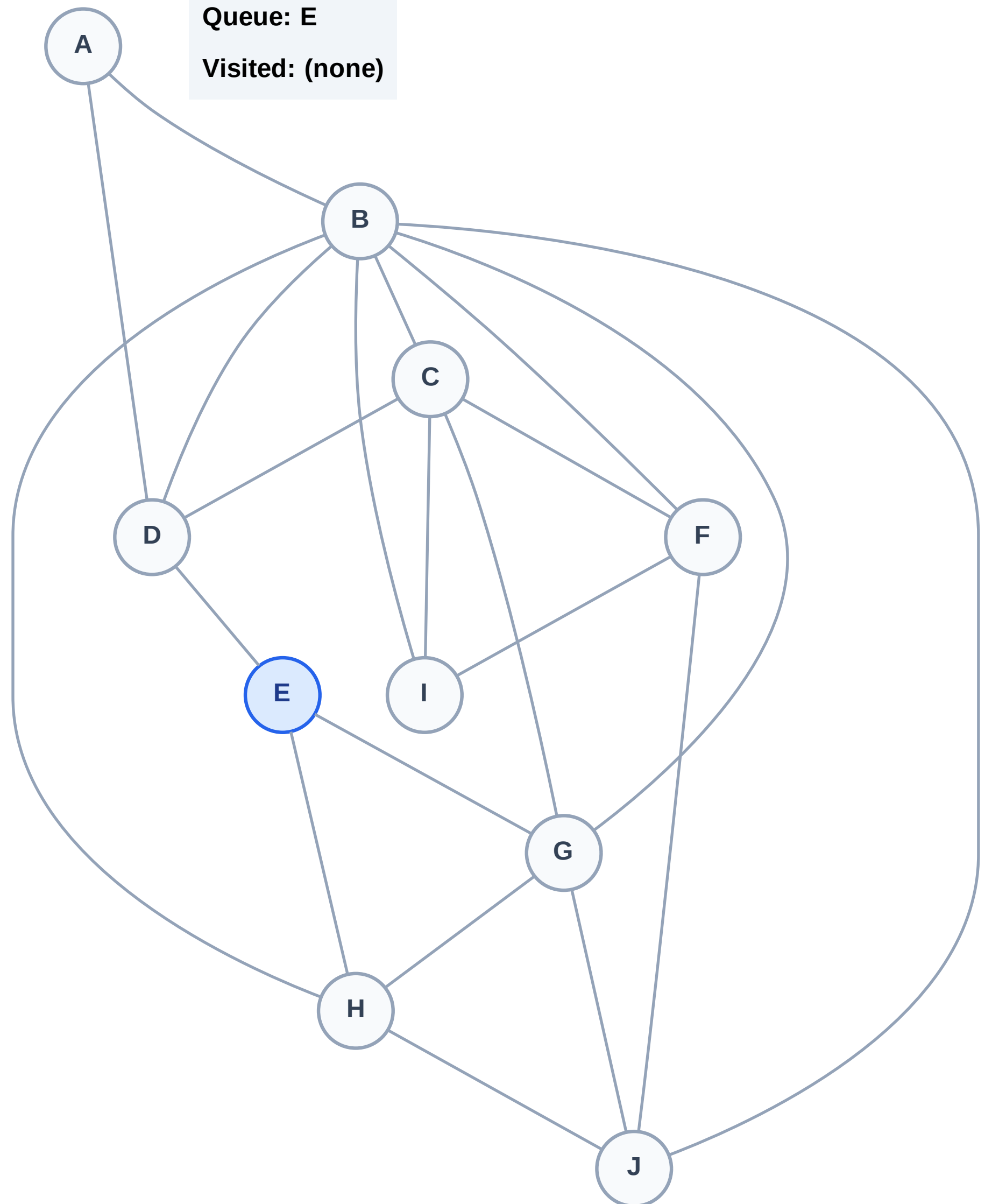
Step 1: Start at E

Legend



Queue: E

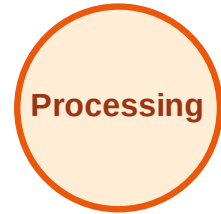
Visited: (none)



BFS Traversal

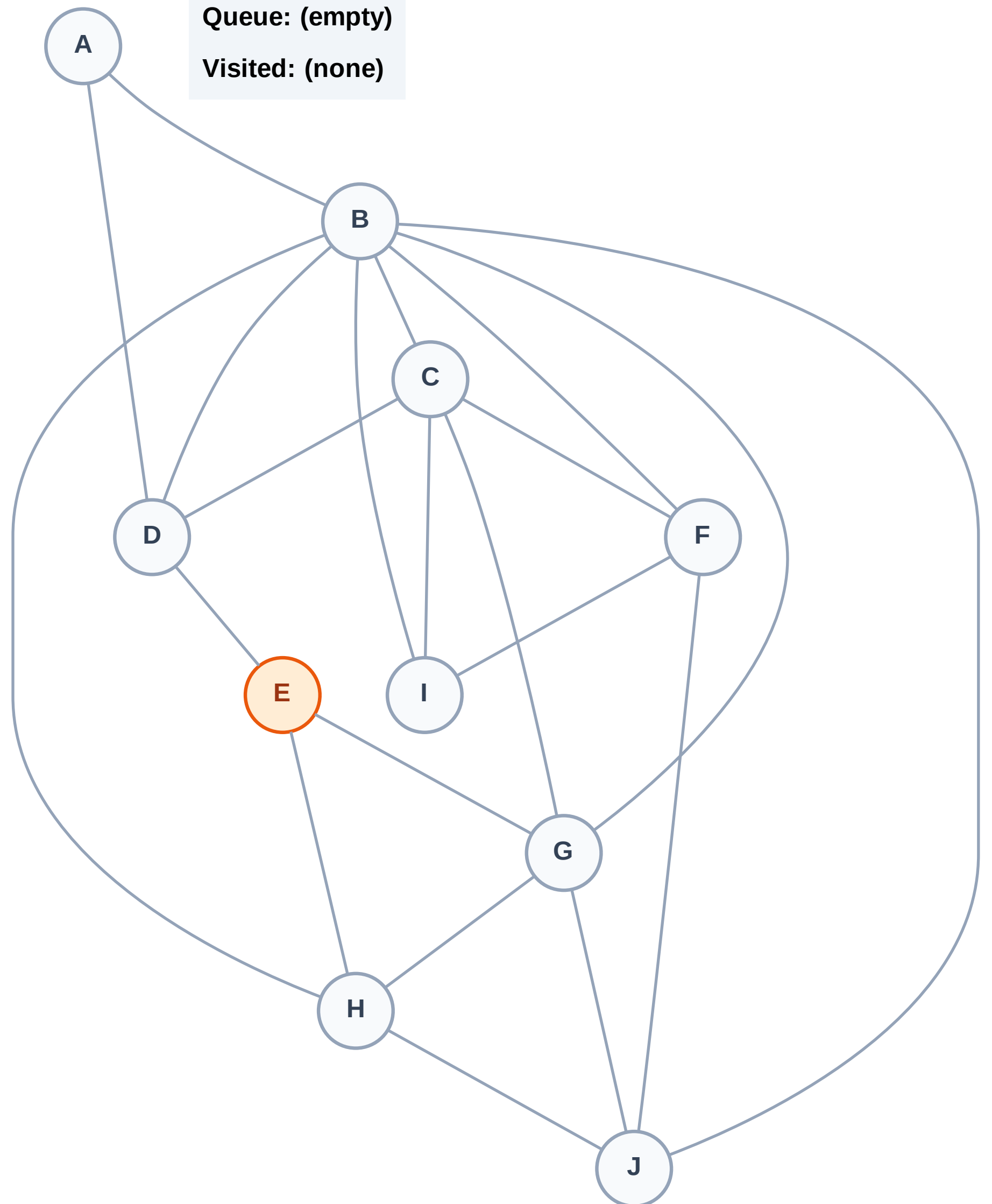
Step 2: Process node E

Legend



Queue: (empty)

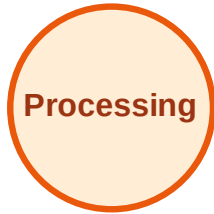
Visited: (none)



BFS Traversal

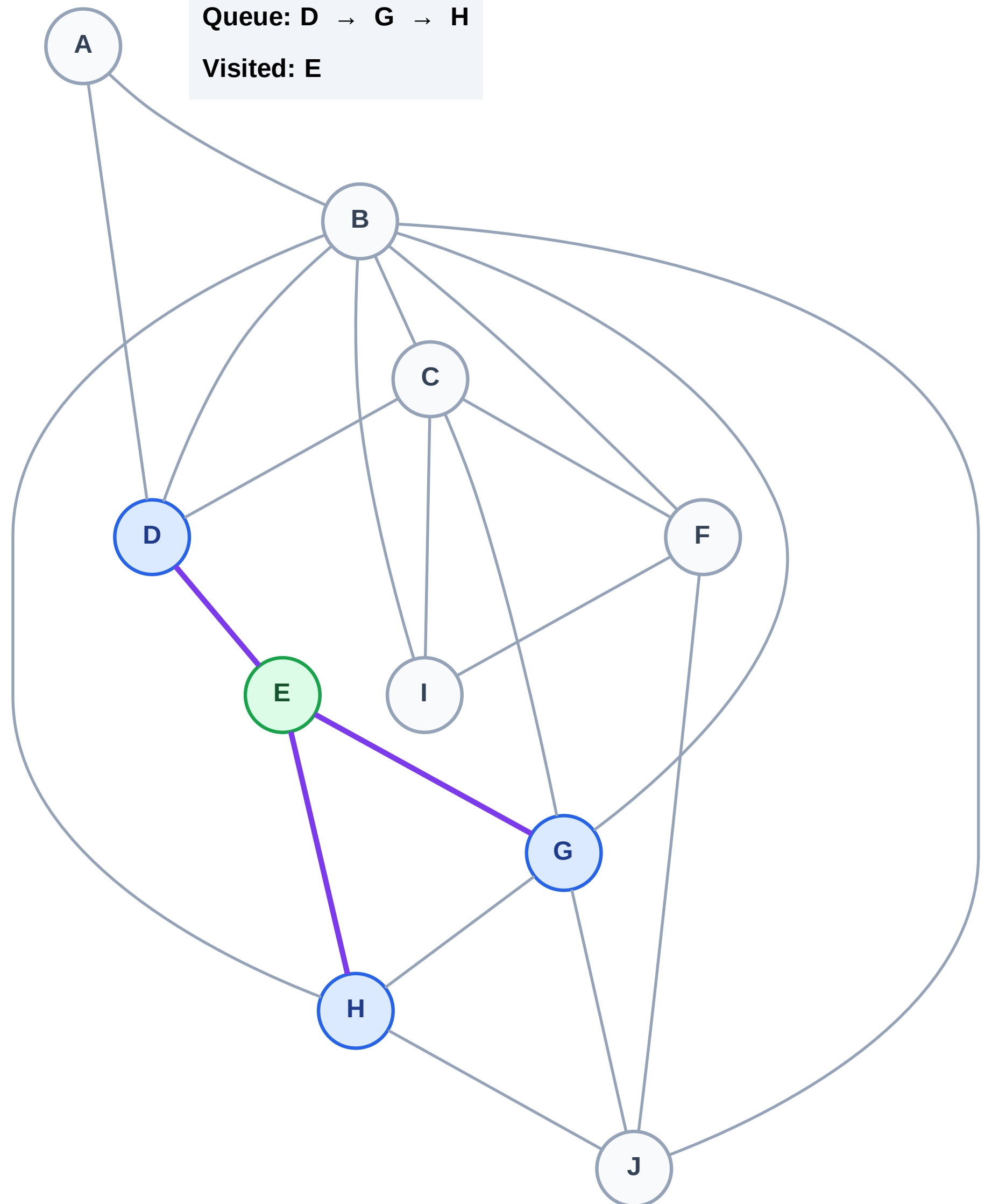
Step 3: Discover D, G, H from E

Legend



Queue: D → G → H

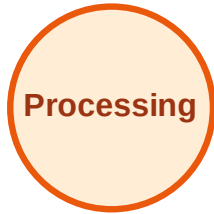
Visited: E



BFS Traversal

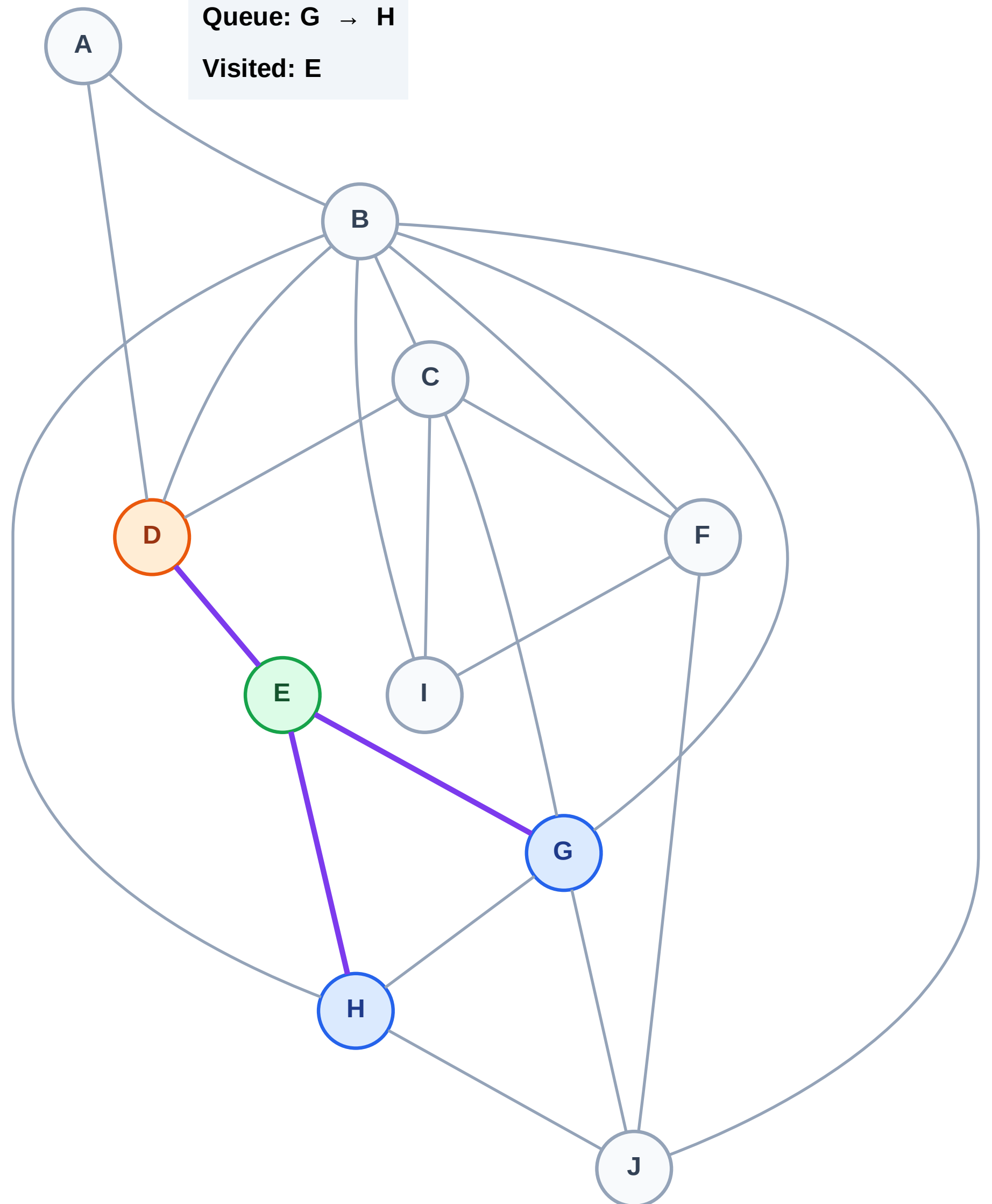
Step 4: Process node D

Legend



Queue: G → H

Visited: E



BFS Traversal

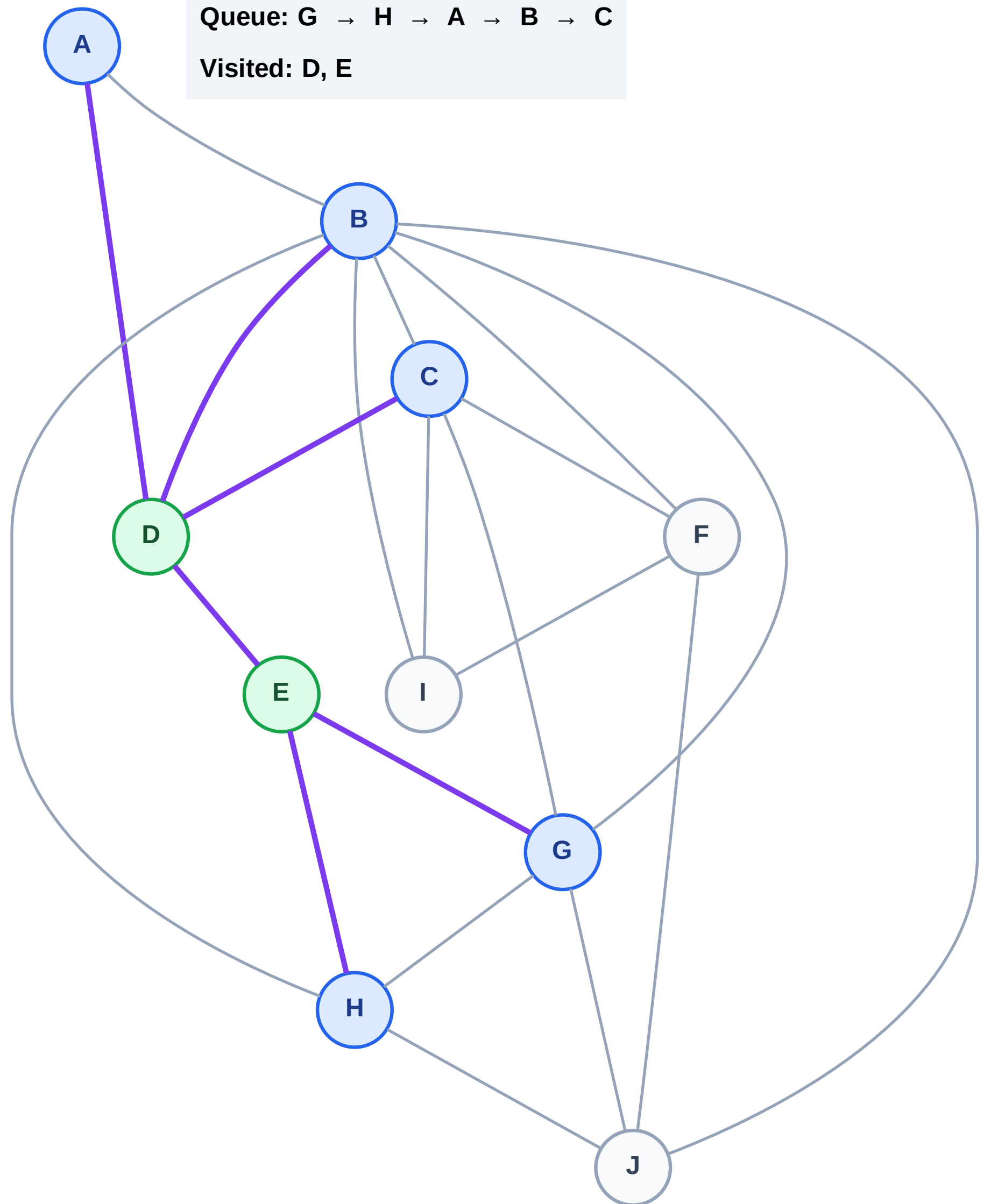
Step 5: Discover A, B, C from D

Legend



Queue: G → H → A → B → C

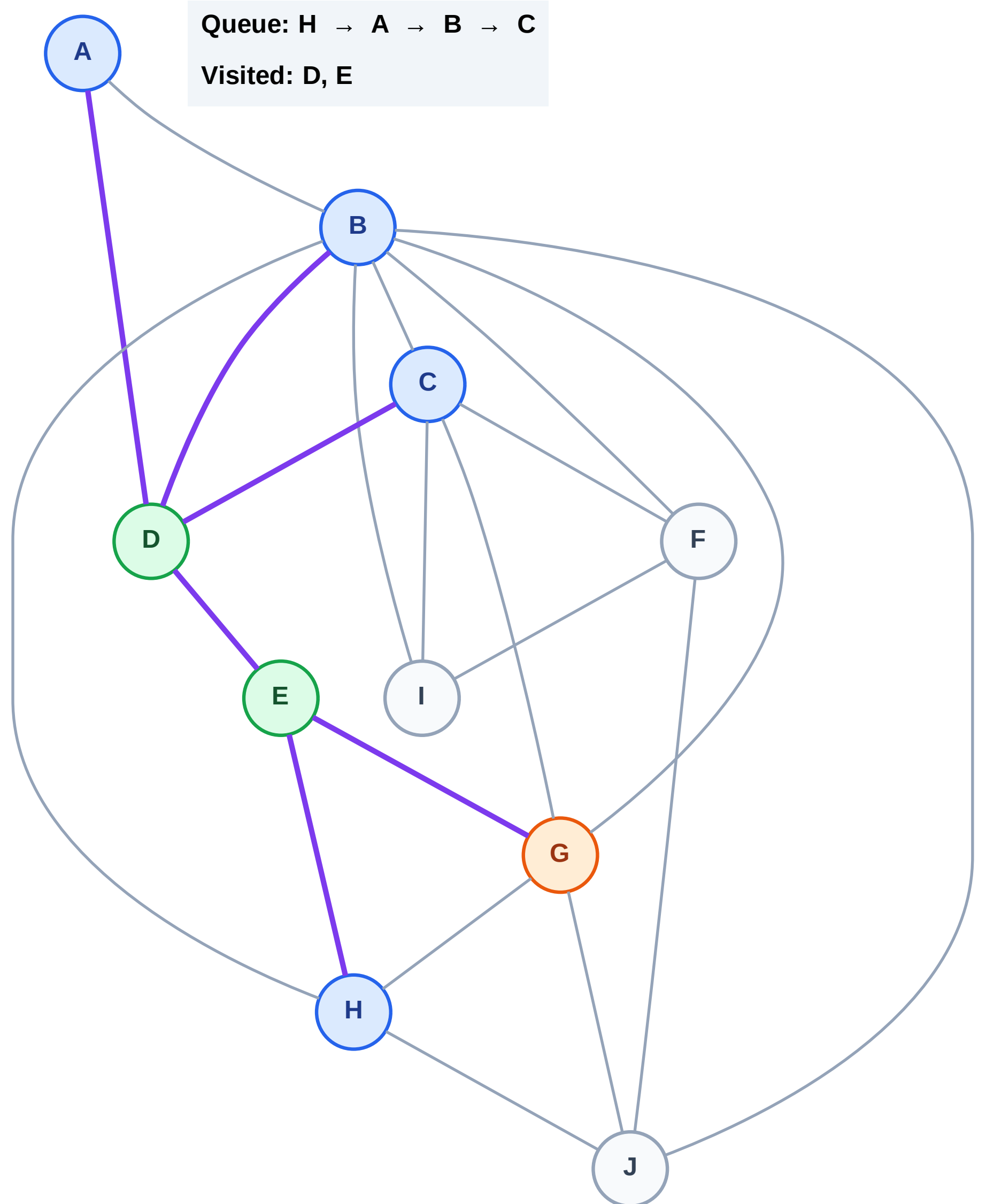
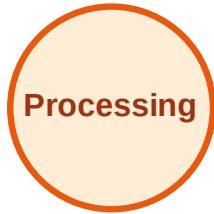
Visited: D, E



BFS Traversal

Step 6: Process node G

Legend



BFS Traversal

Step 7: Discover J from G

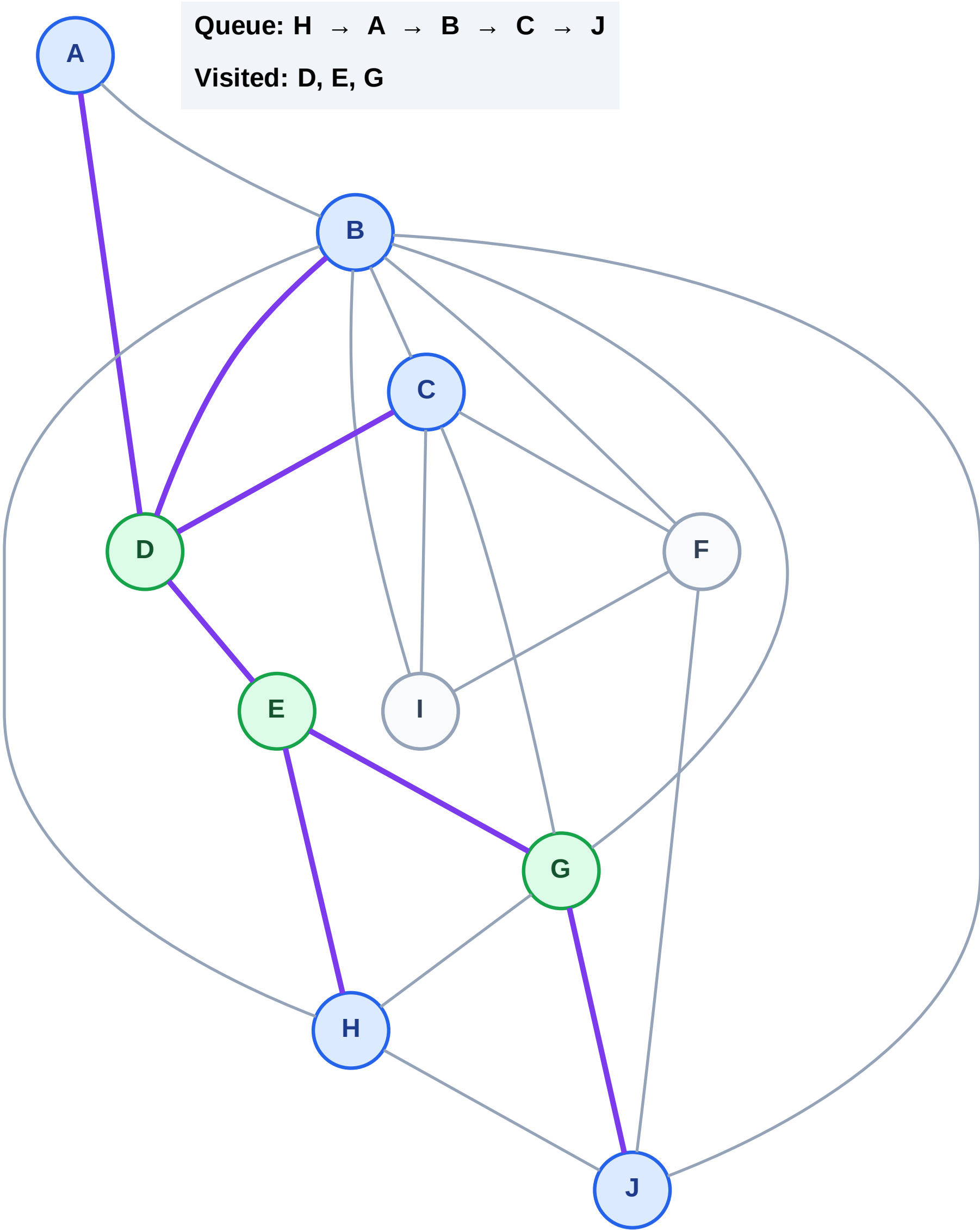
Legend

Complete

Processing

Discovered

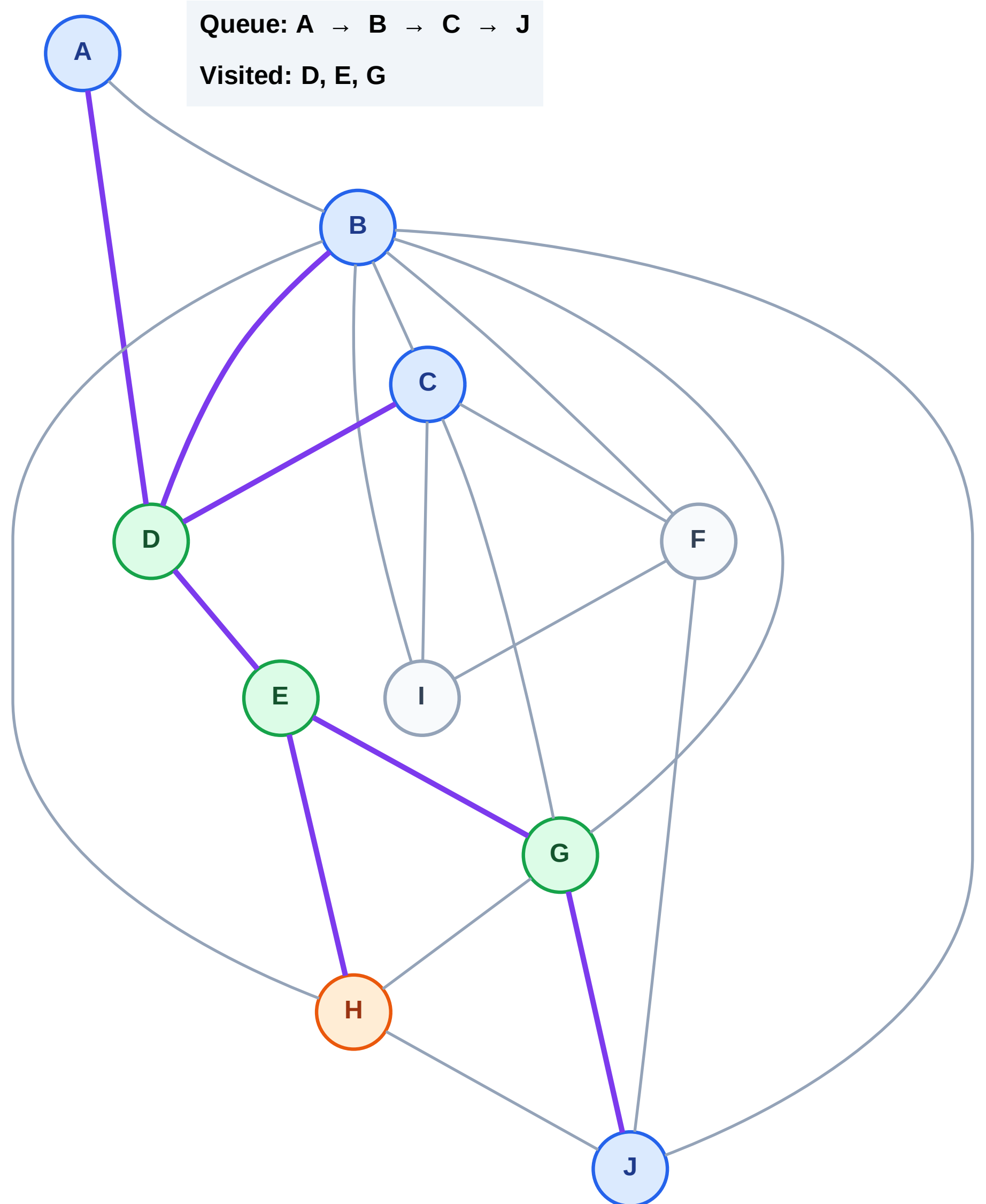
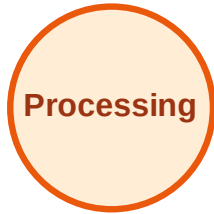
Unseen



BFS Traversal

Step 8: Process node H

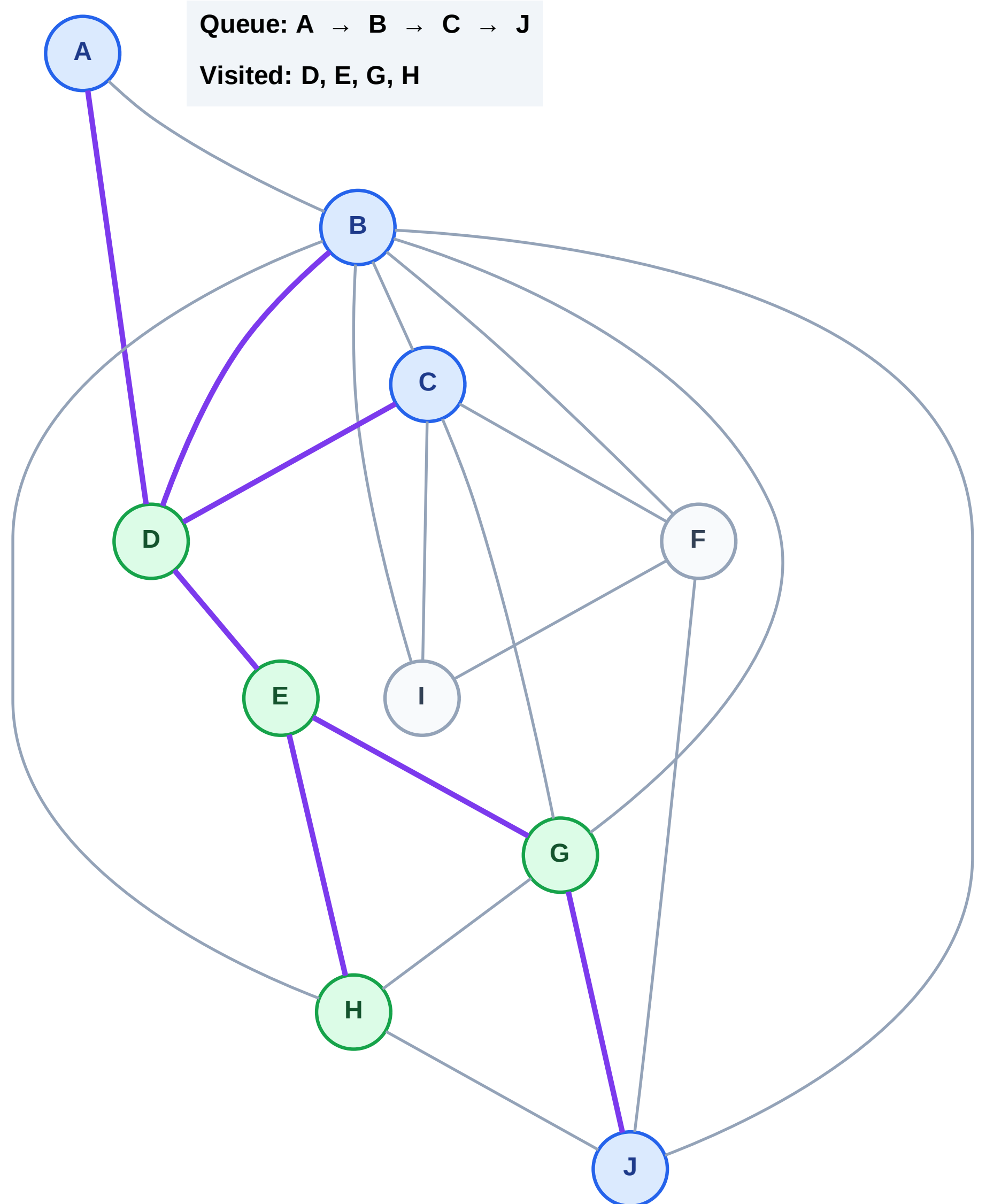
Legend



BFS Traversal

Step 9: No new neighbors from H

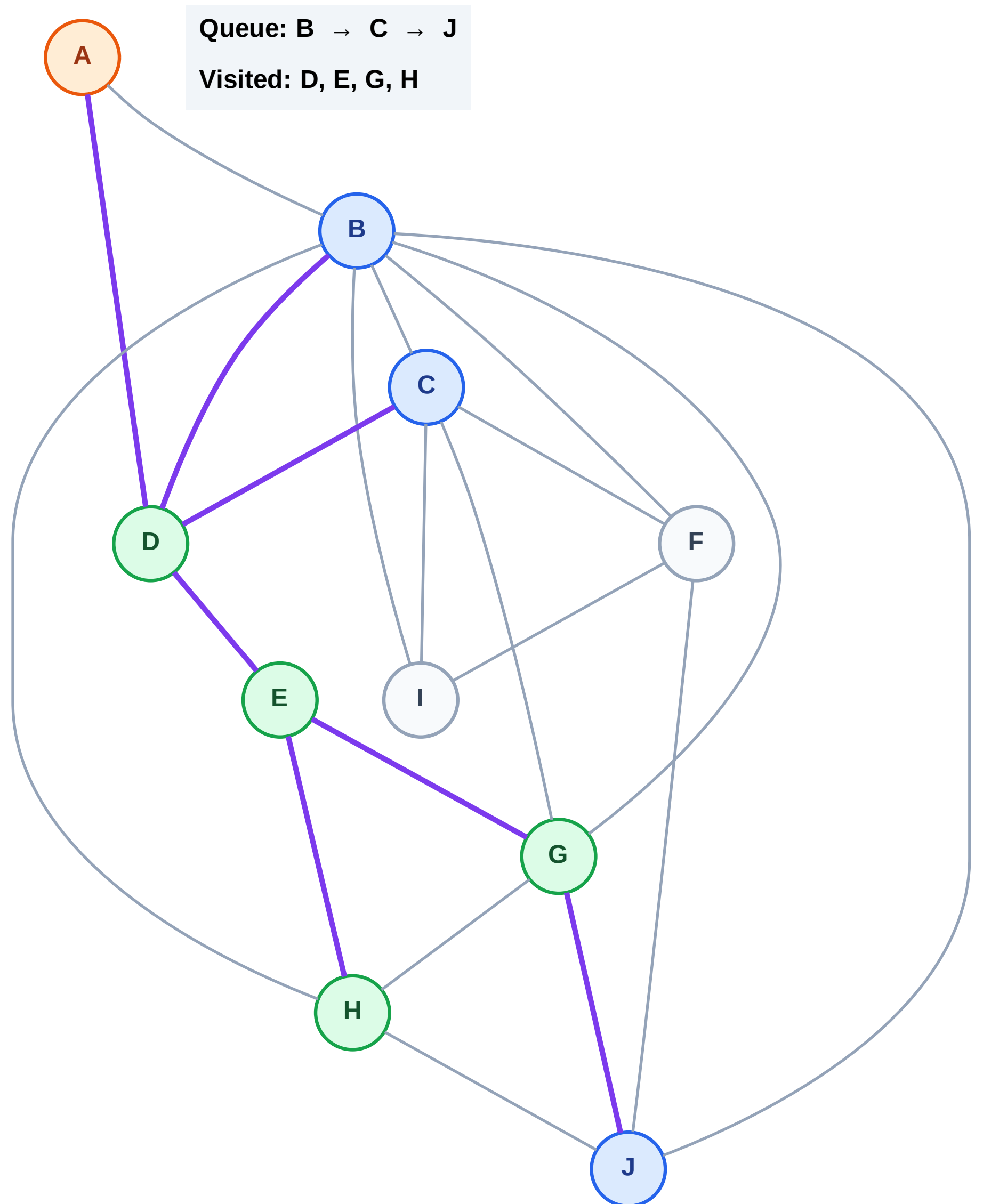
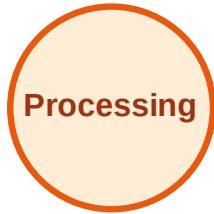
Legend



BFS Traversal

Step 10: Process node A

Legend



BFS Traversal

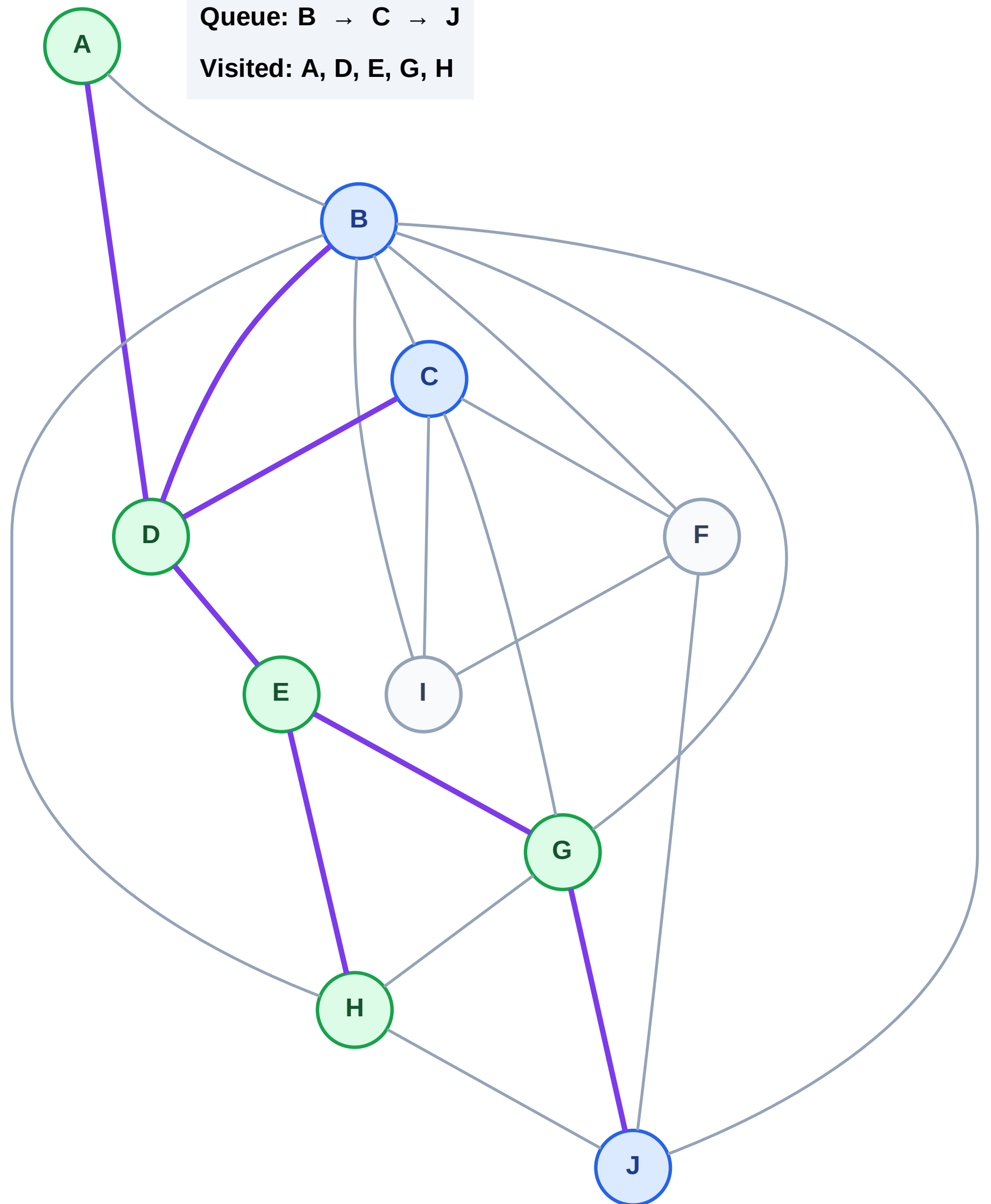
Step 11: No new neighbors from A

Legend



Queue: B → C → J

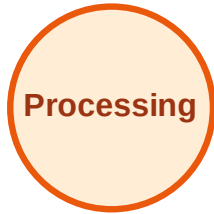
Visited: A, D, E, G, H



BFS Traversal

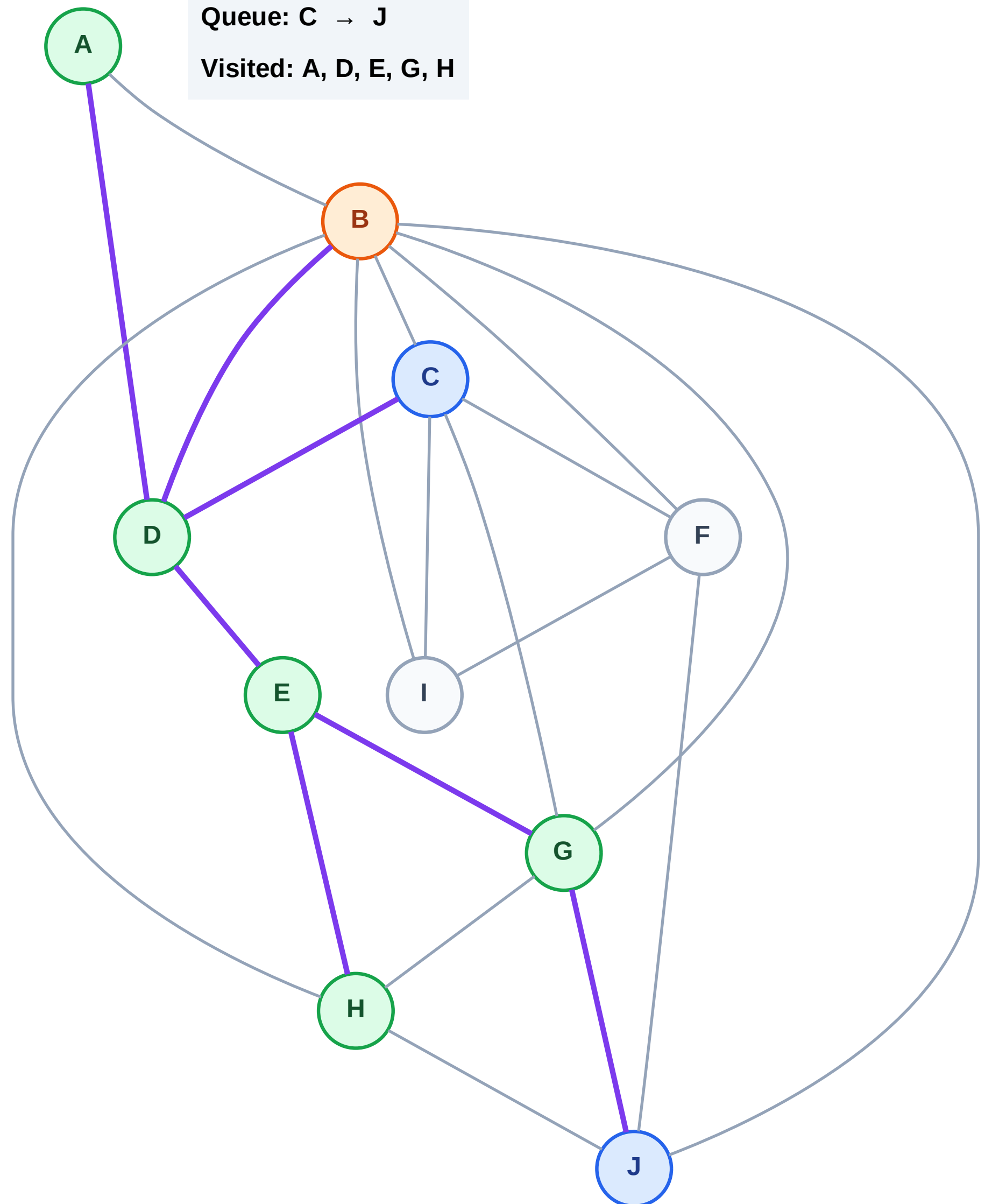
Step 12: Process node B

Legend



Queue: C → J

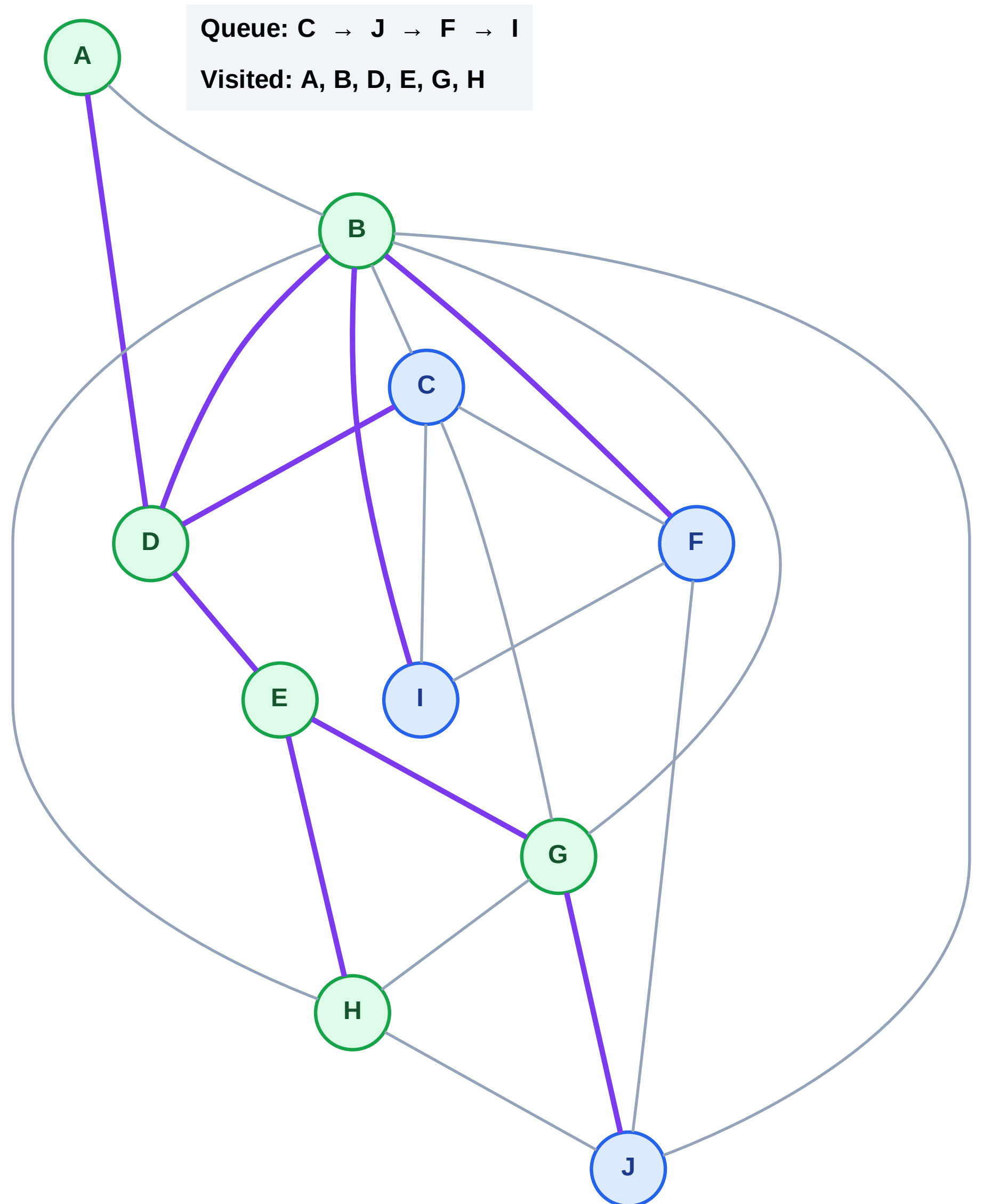
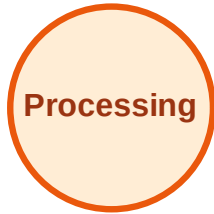
Visited: A, D, E, G, H



BFS Traversal

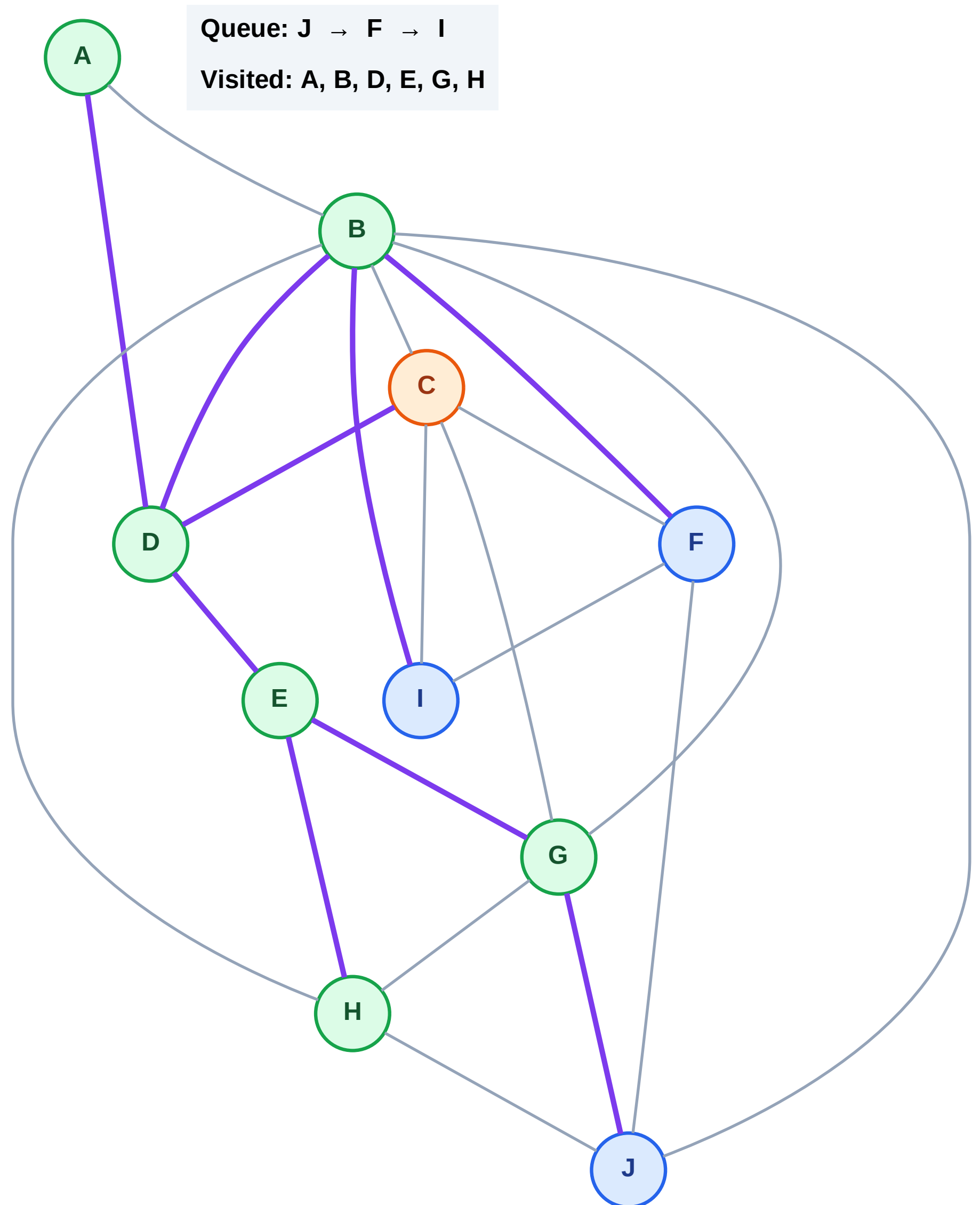
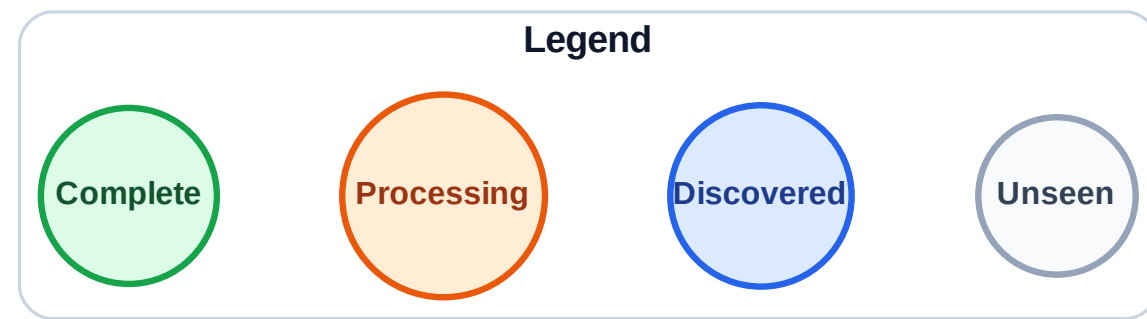
Step 13: Discover F, I from B

Legend



BFS Traversal

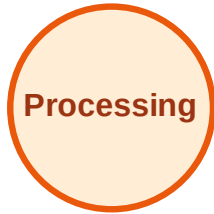
Step 14: Process node C



BFS Traversal

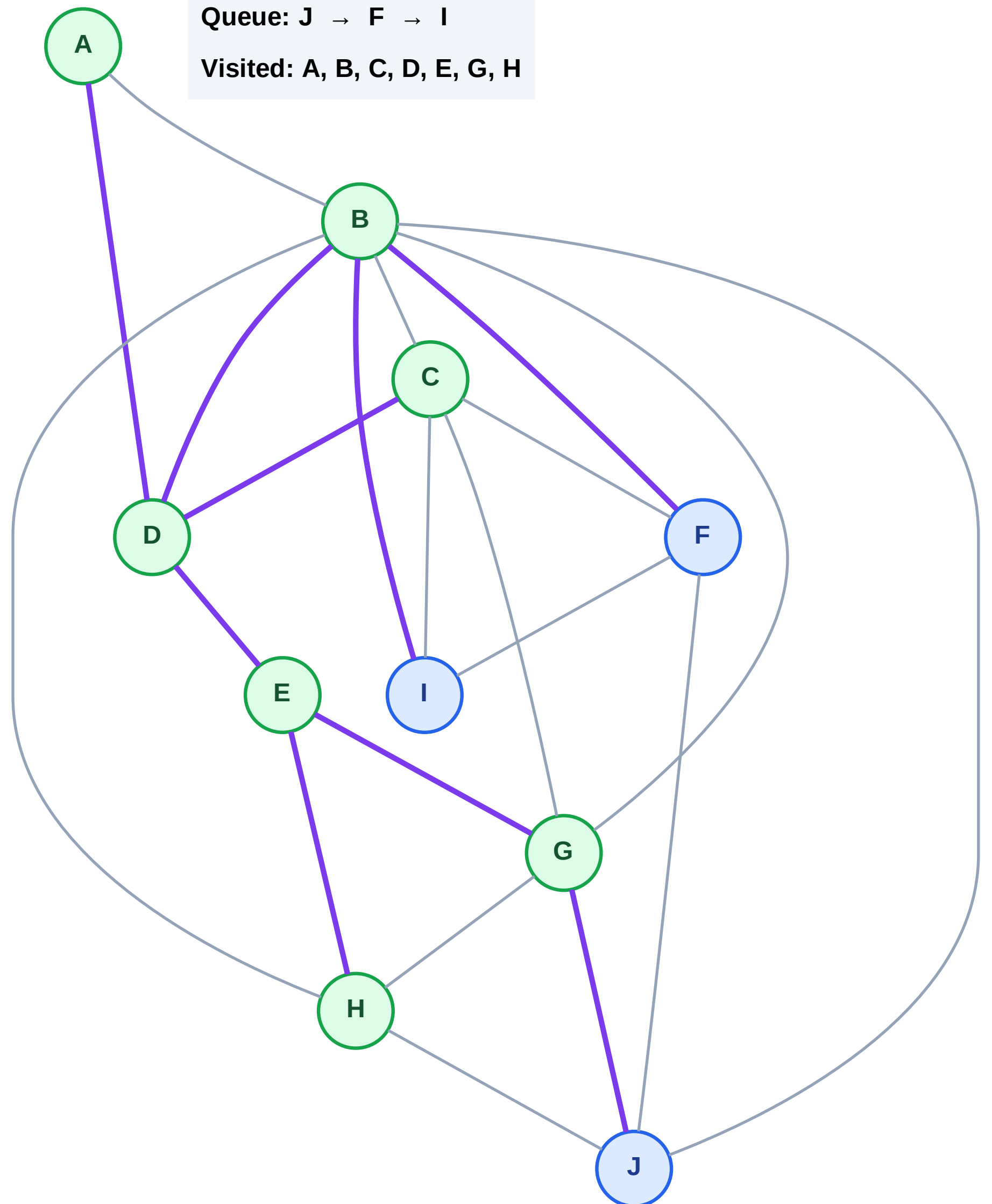
Step 15: No new neighbors from C

Legend



Queue: J → F → I

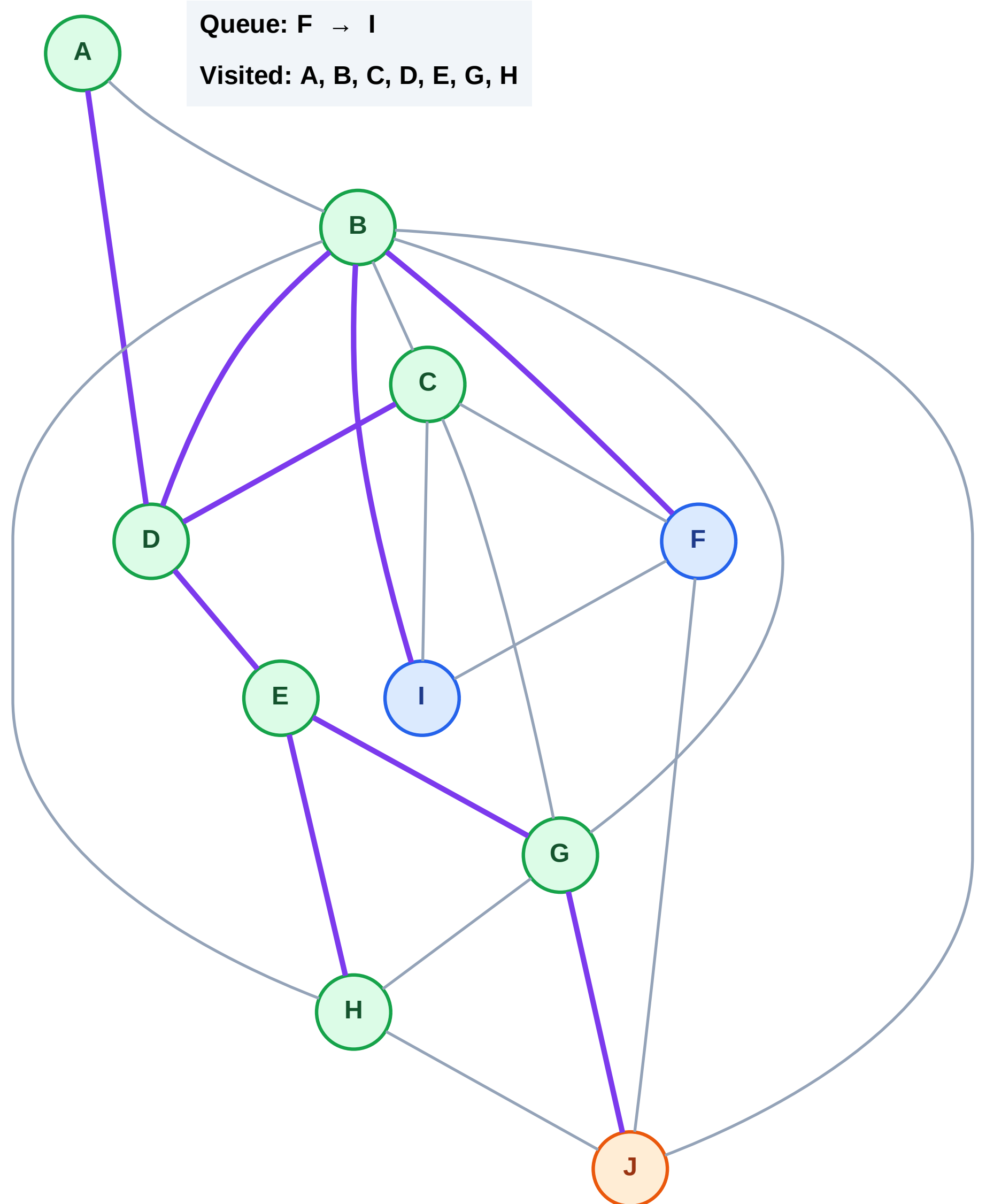
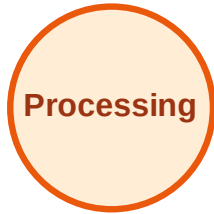
Visited: A, B, C, D, E, G, H



BFS Traversal

Step 16: Process node J

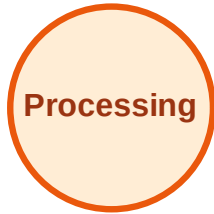
Legend



BFS Traversal

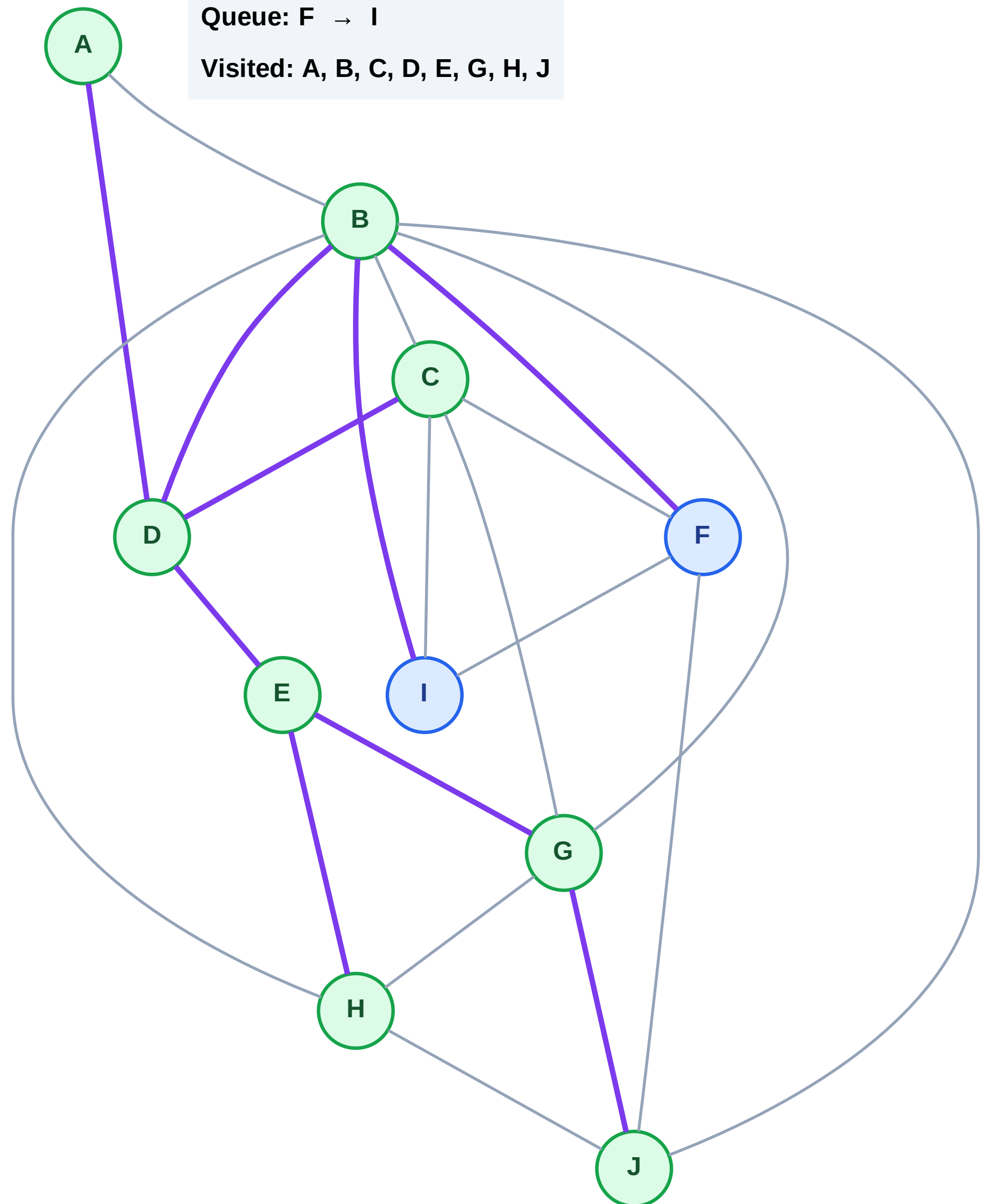
Step 17: No new neighbors from J

Legend



Queue: F → I

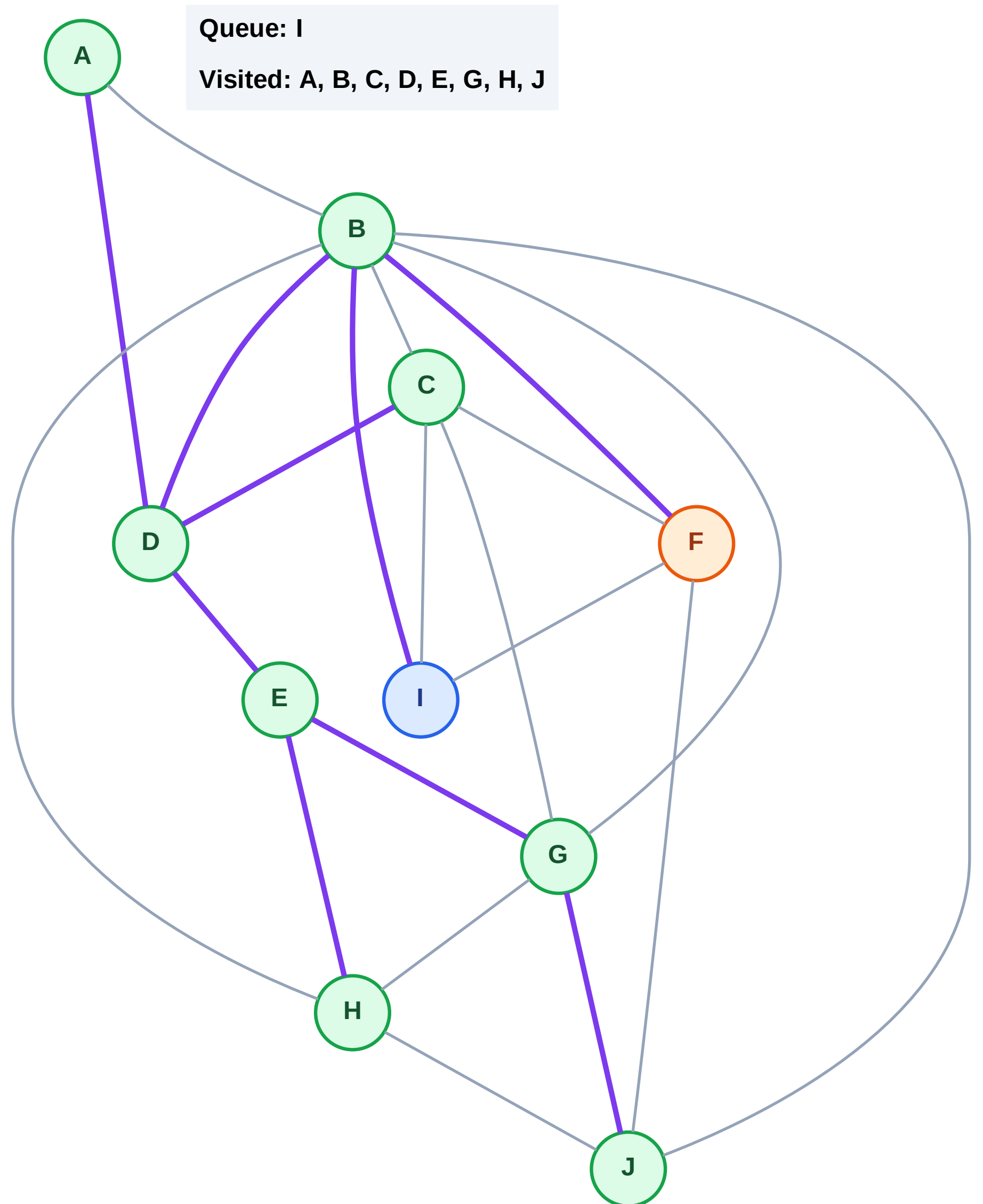
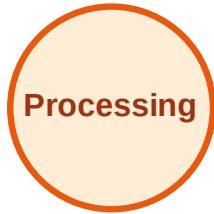
Visited: A, B, C, D, E, G, H, J



BFS Traversal

Step 18: Process node F

Legend



BFS Traversal

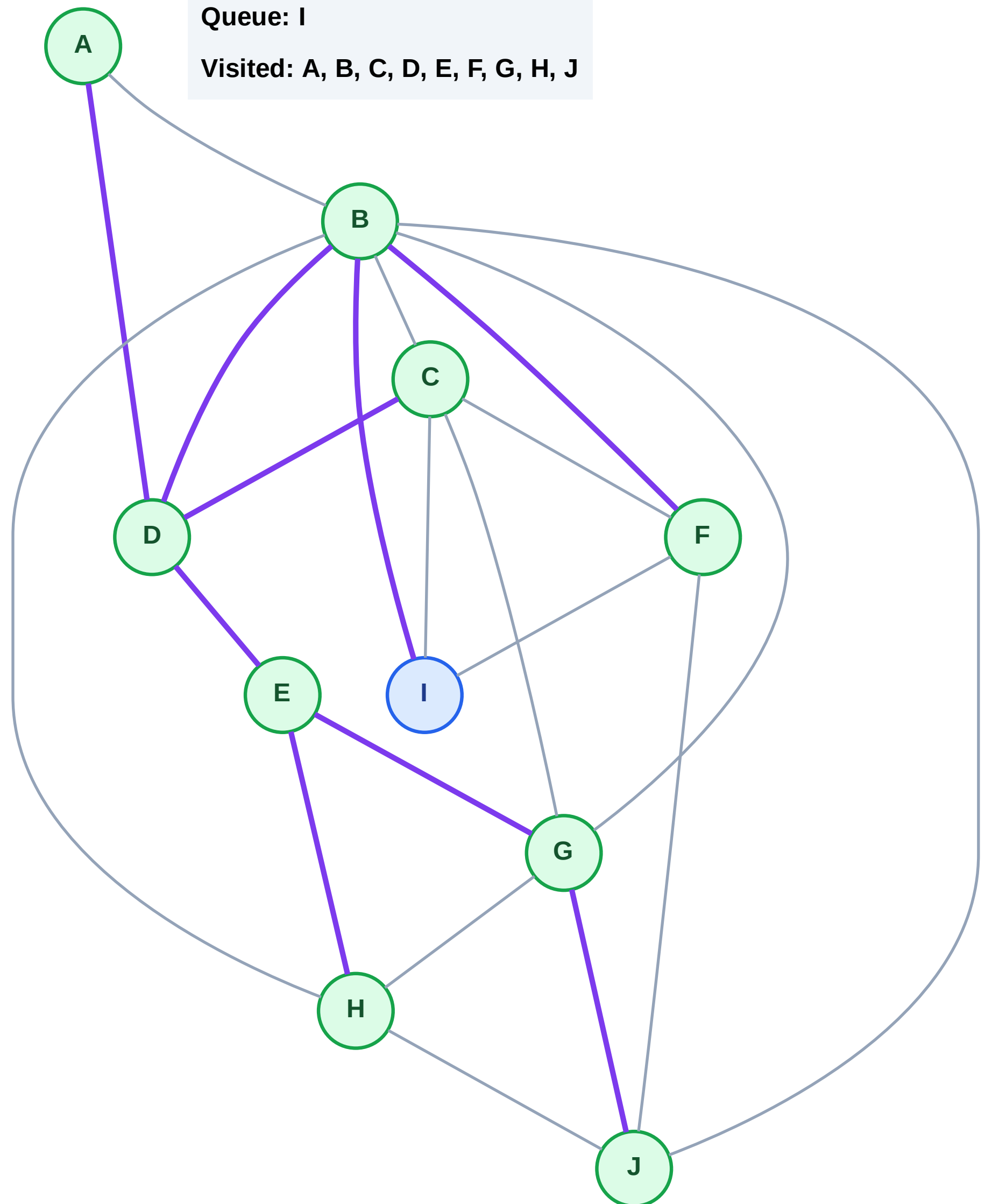
Step 19: No new neighbors from F

Legend



Queue: I

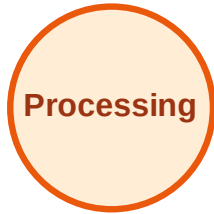
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

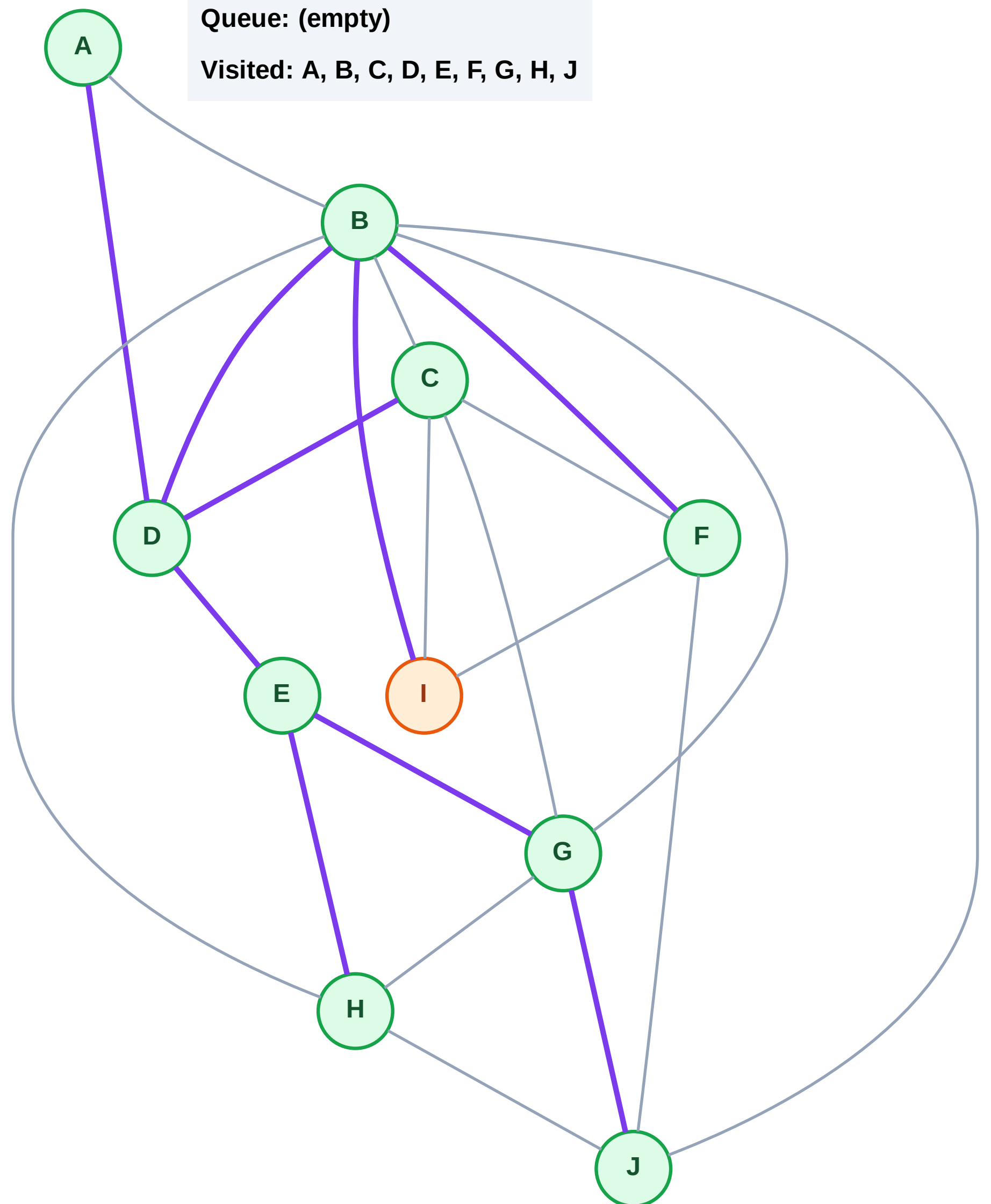
Step 20: Process node I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

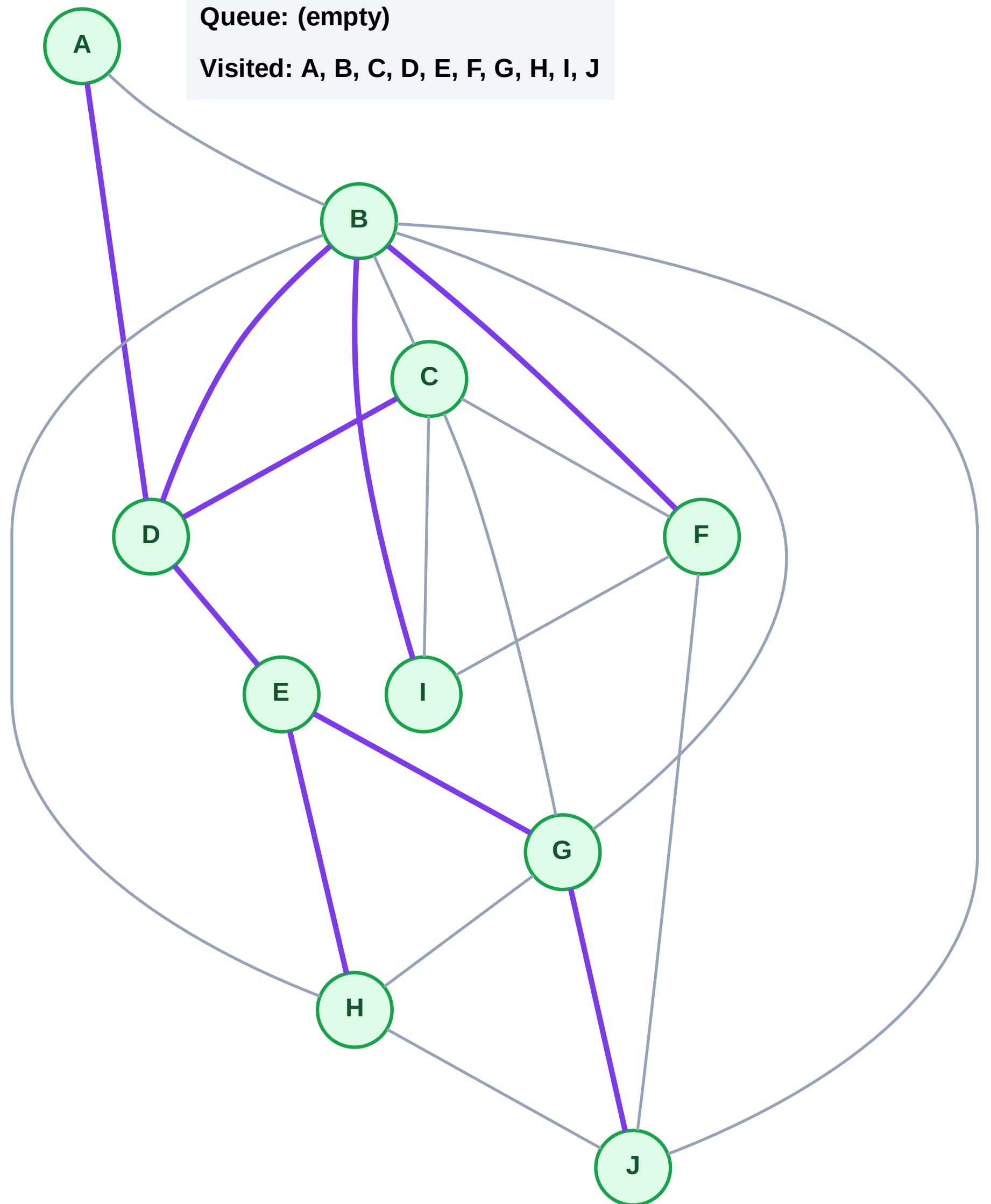
Step 21: No new neighbors from I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



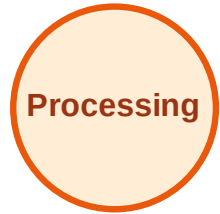
DFS Traversal

Step 1: Start at E

Legend



Complete



Processing



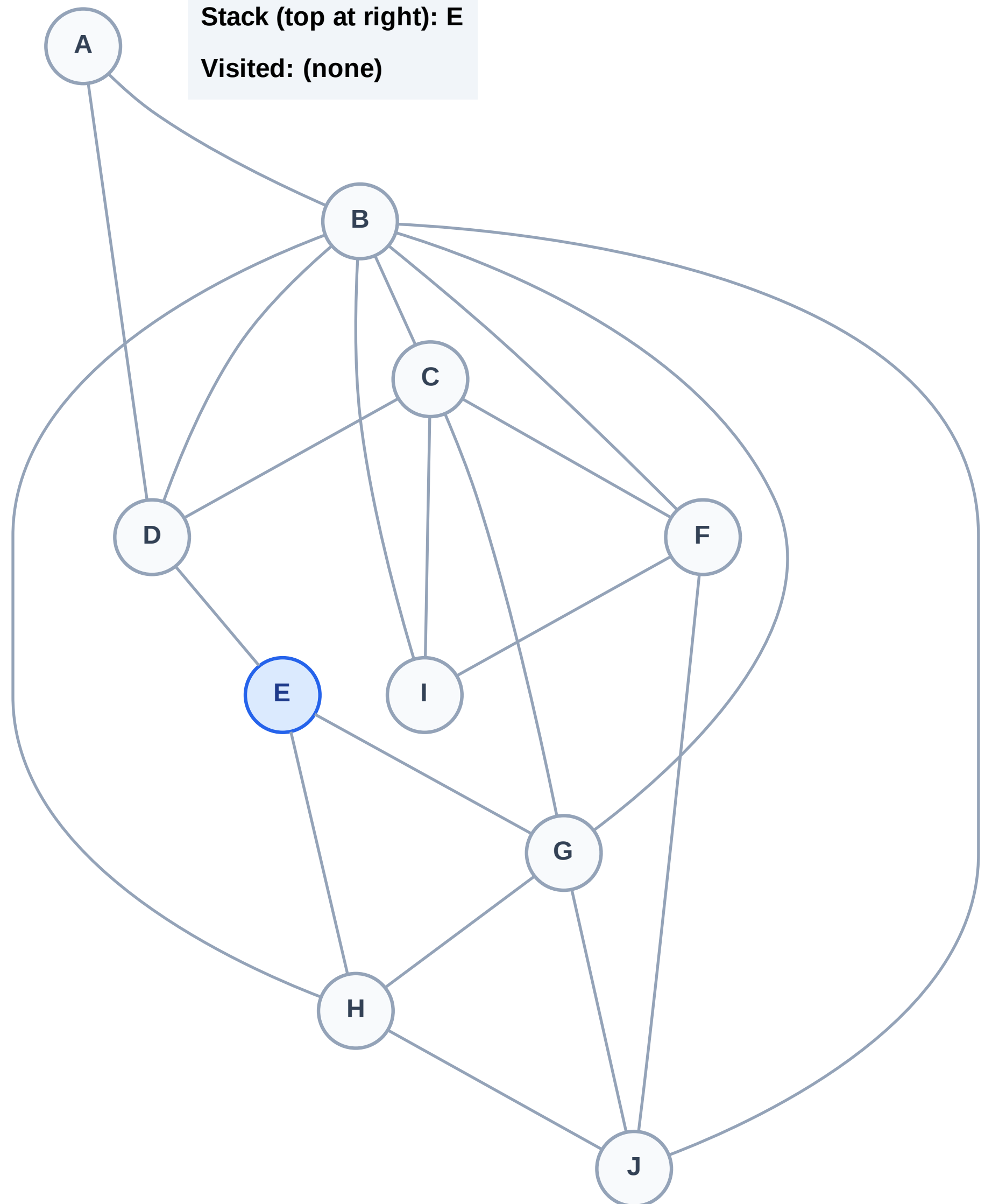
Discovered



Unseen

Stack (top at right): E

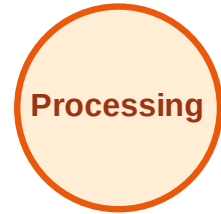
Visited: (none)



DFS Traversal

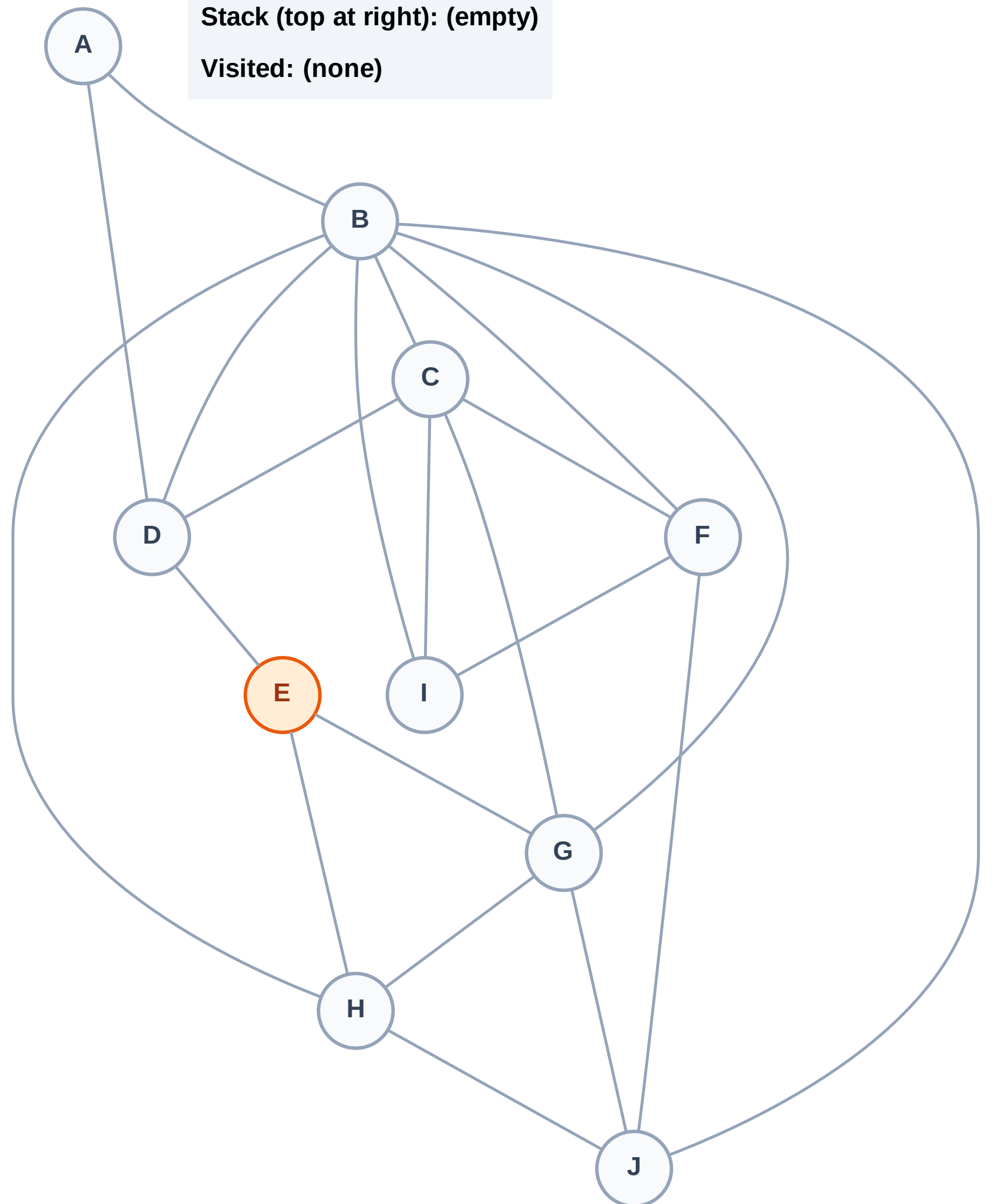
Step 2: Process node E

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

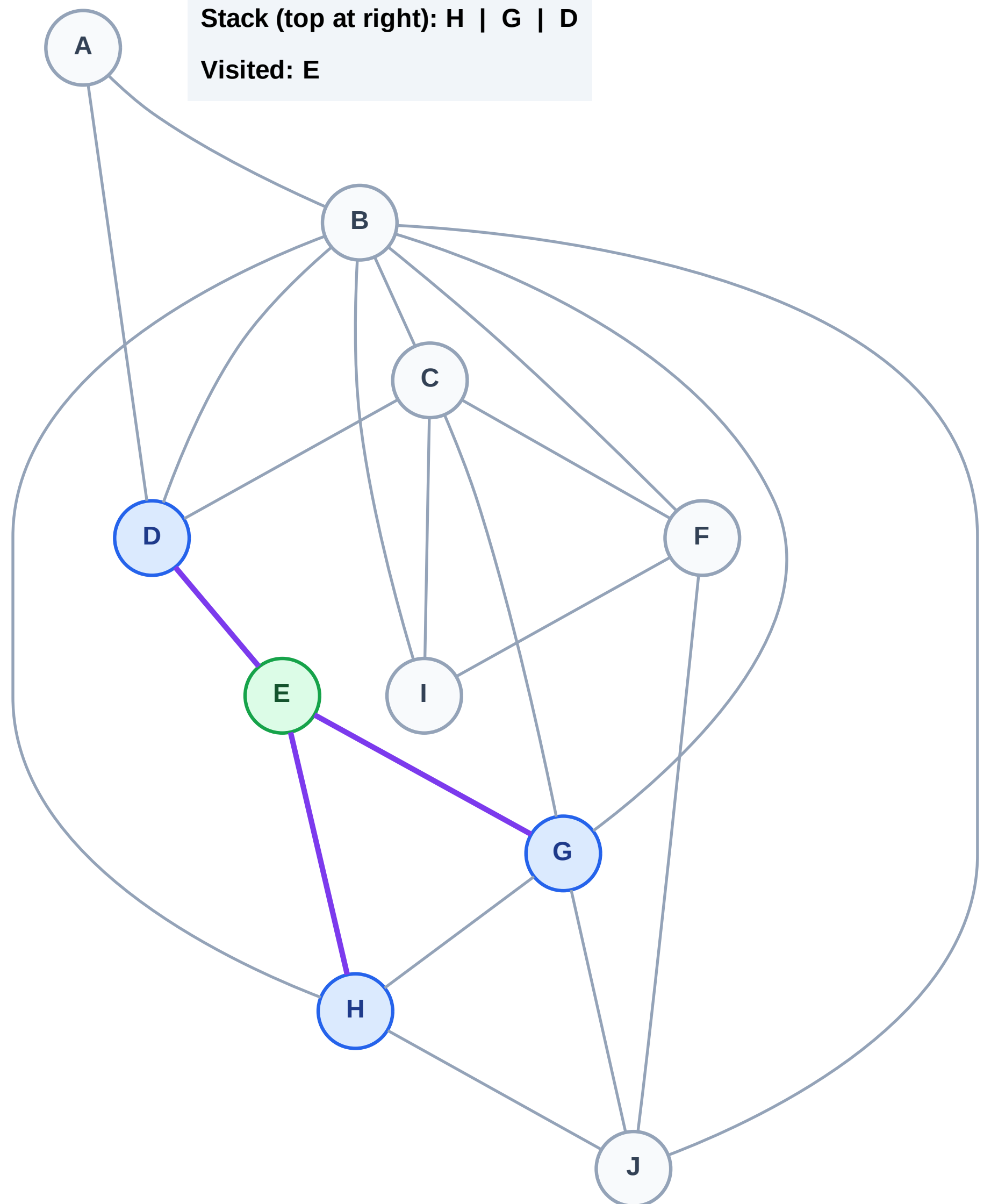
Step 3: Discover H, G, D from E

Legend



Stack (top at right): H | G | D

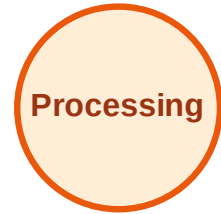
Visited: E



DFS Traversal

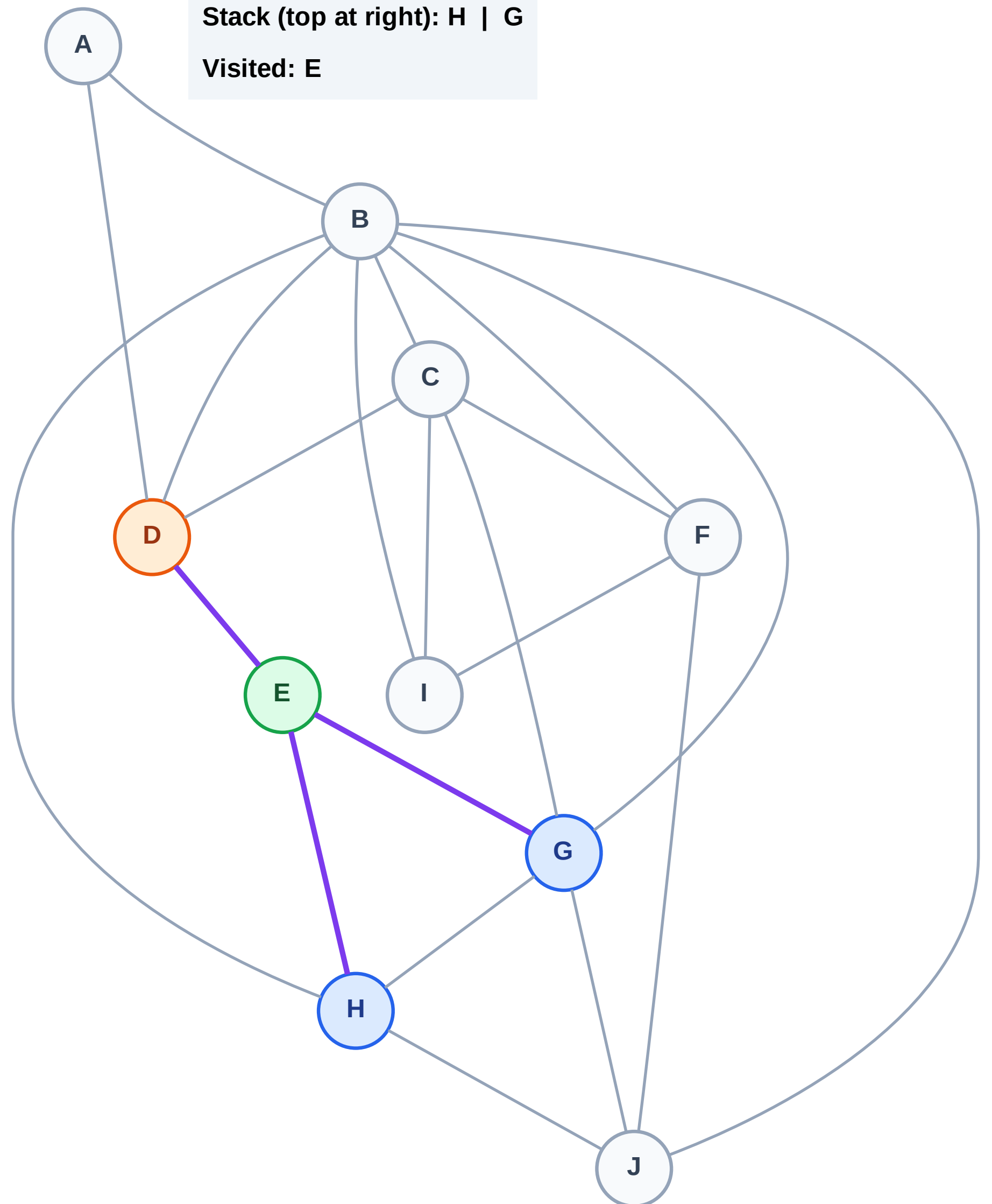
Step 4: Process node D

Legend



Stack (top at right): H | G

Visited: E



DFS Traversal

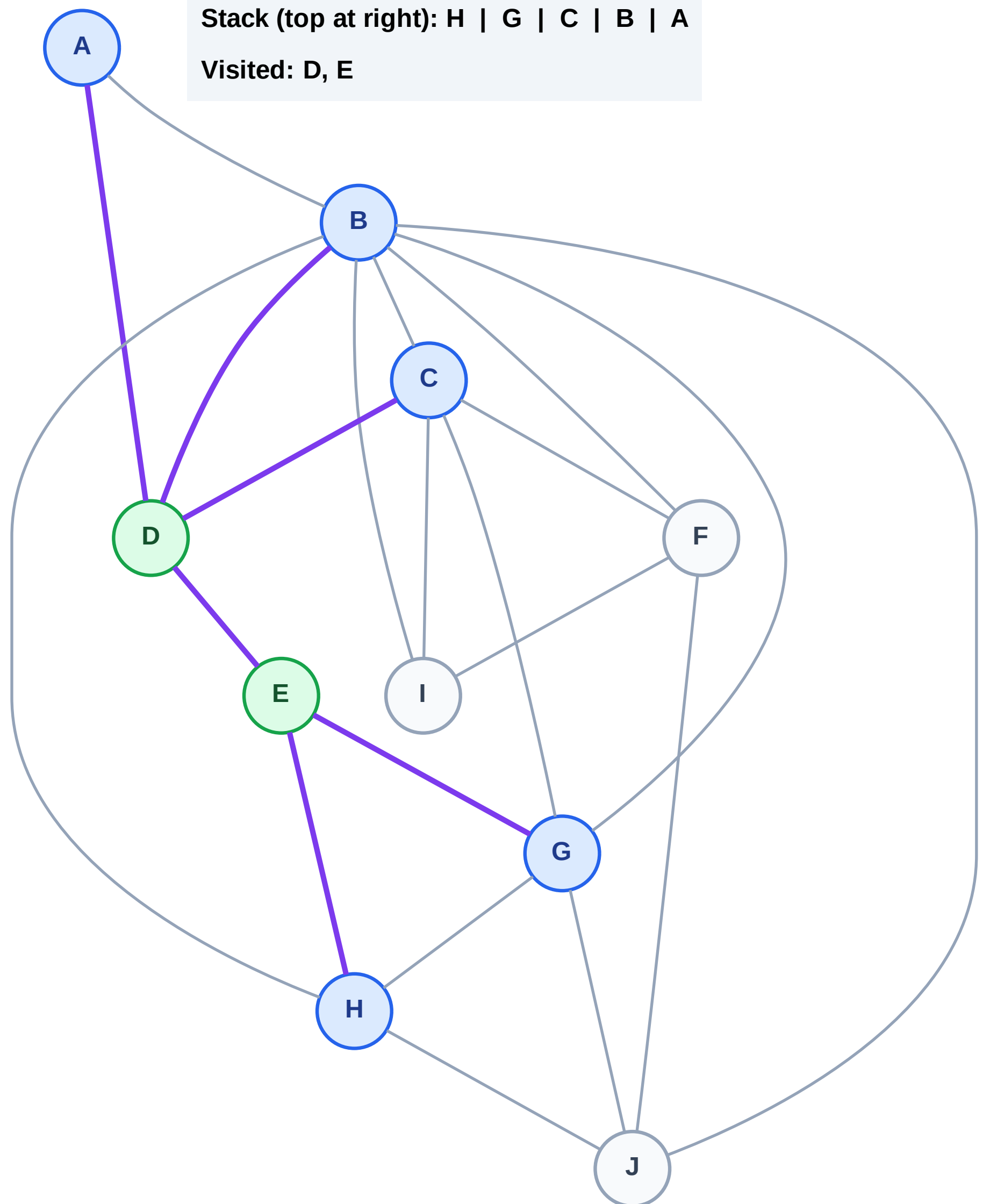
Step 5: Discover C, B, A from D

Legend



Stack (top at right): H | G | C | B | A

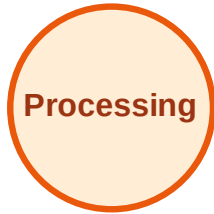
Visited: D, E



DFS Traversal

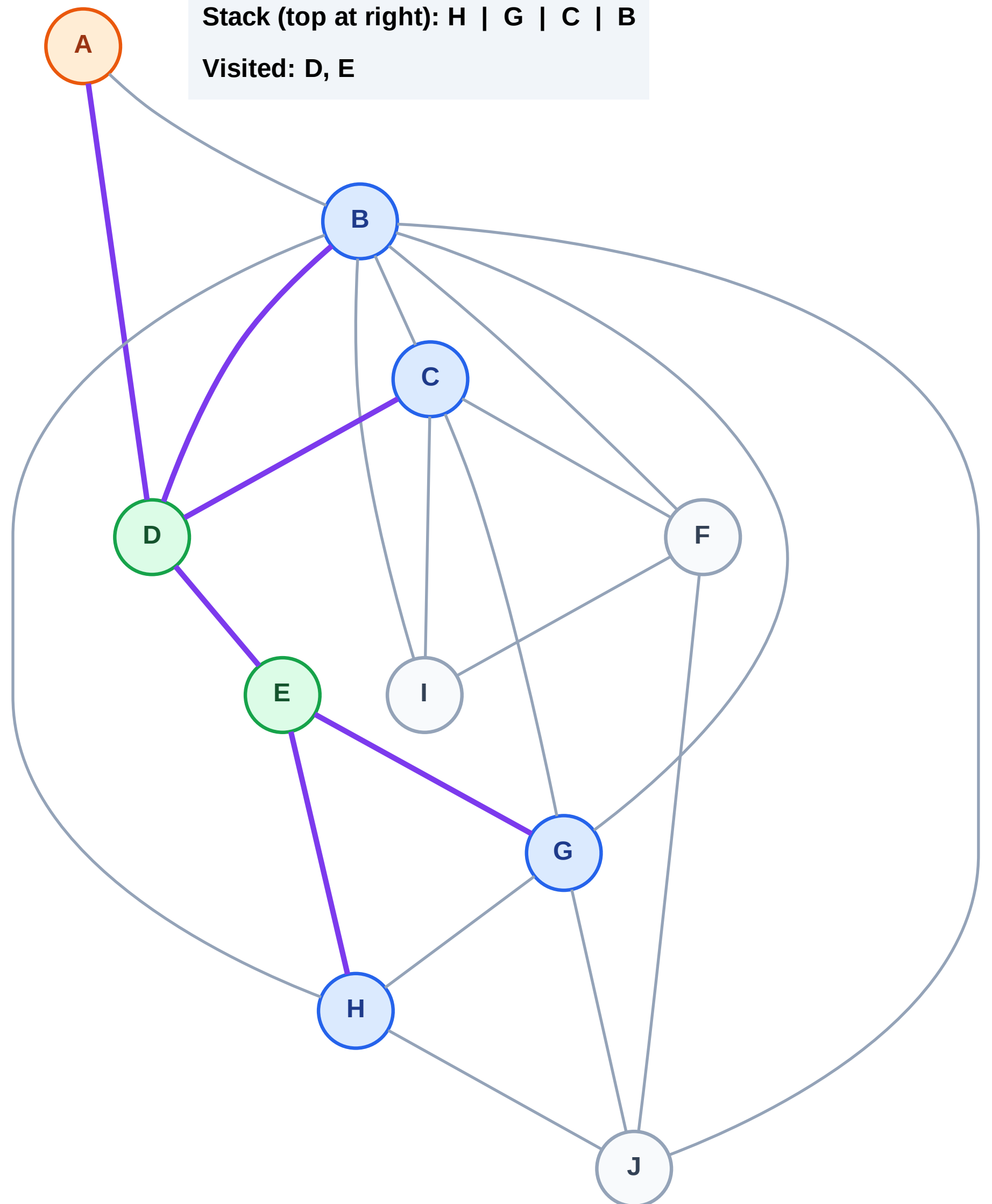
Step 6: Process node A

Legend



Stack (top at right): H | G | C | B

Visited: D, E



DFS Traversal

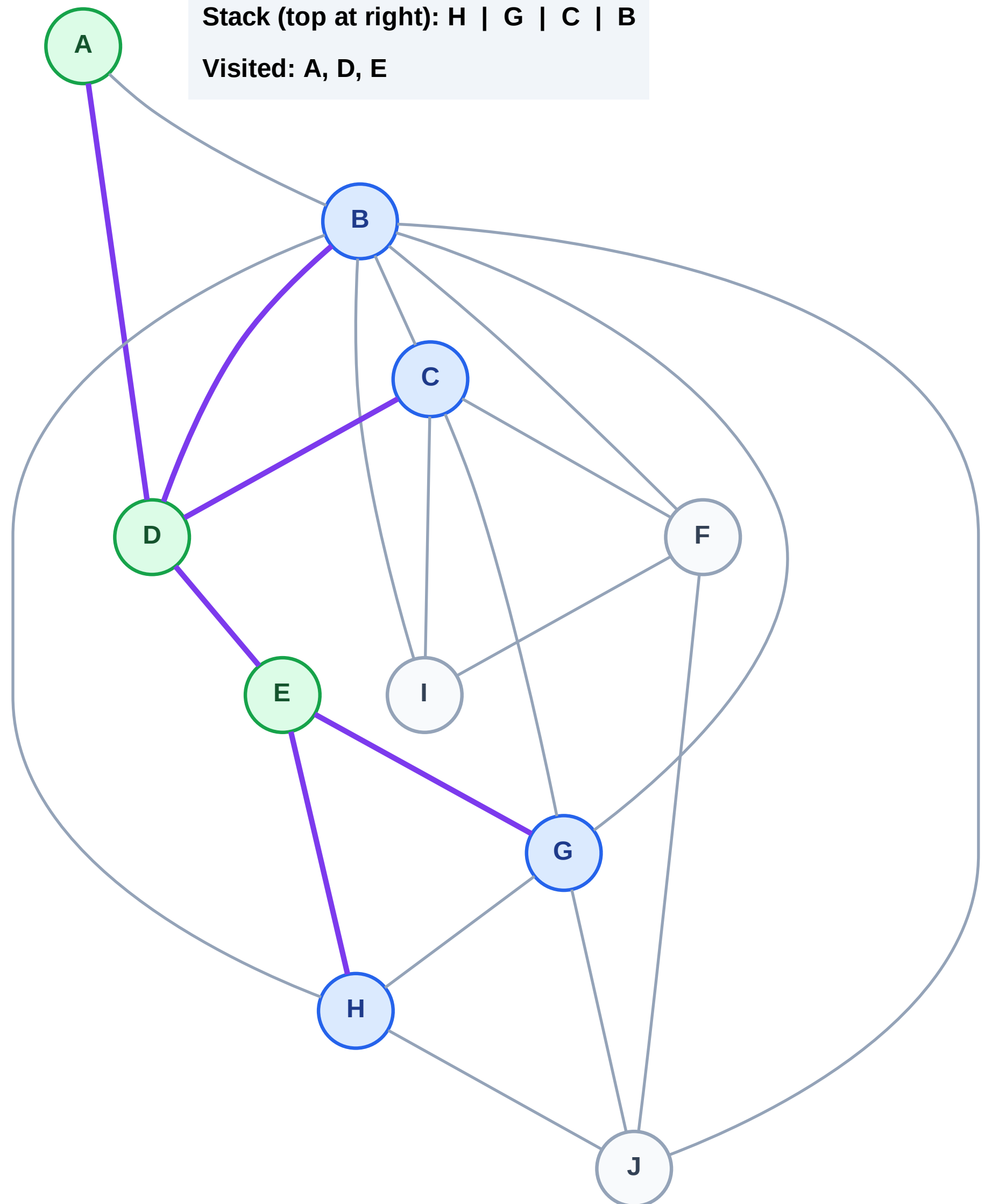
Step 7: No new neighbors from A

Legend



Stack (top at right): H | G | C | B

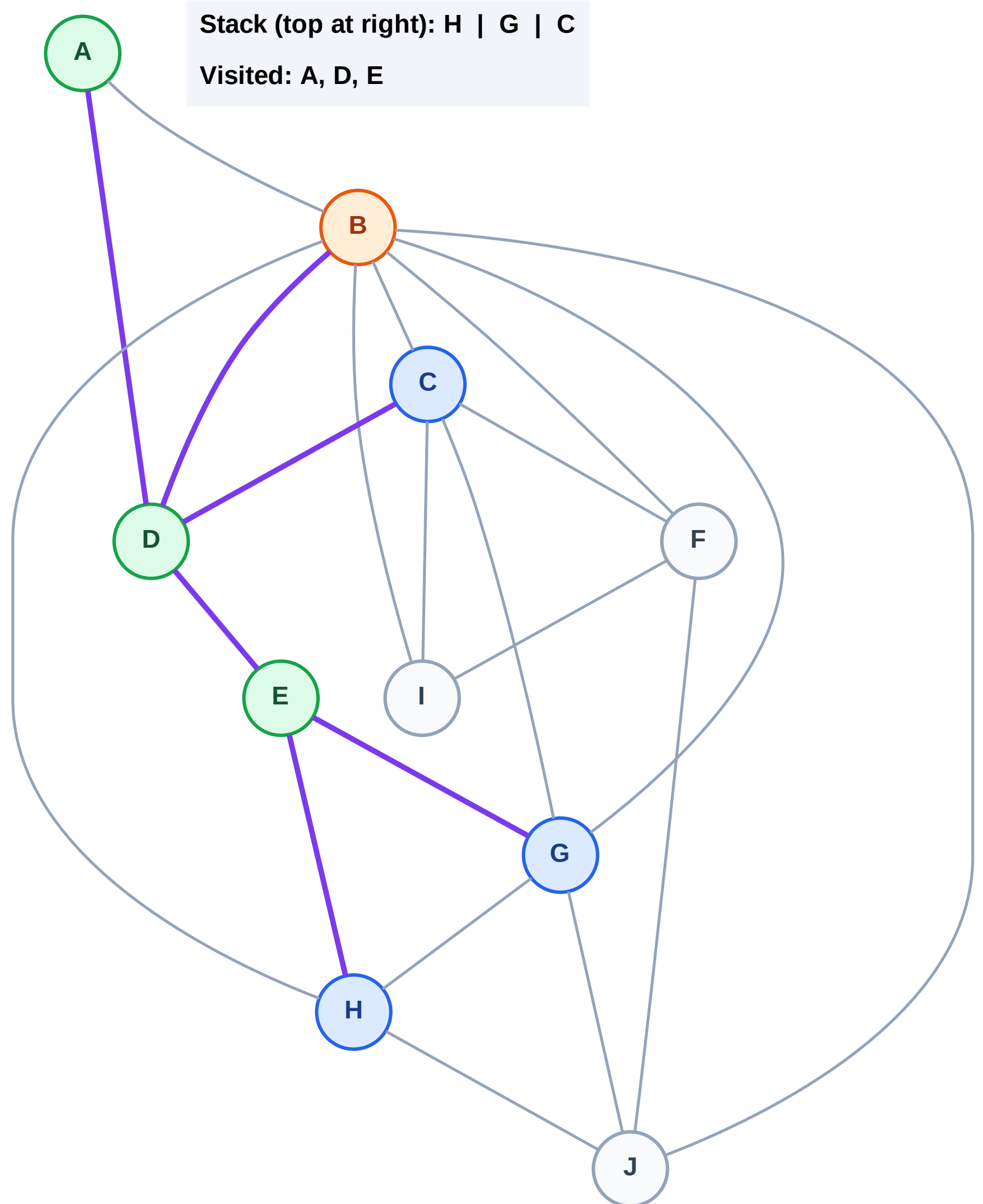
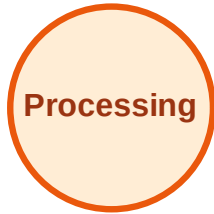
Visited: A, D, E



DFS Traversal

Step 8: Process node B

Legend



DFS Traversal

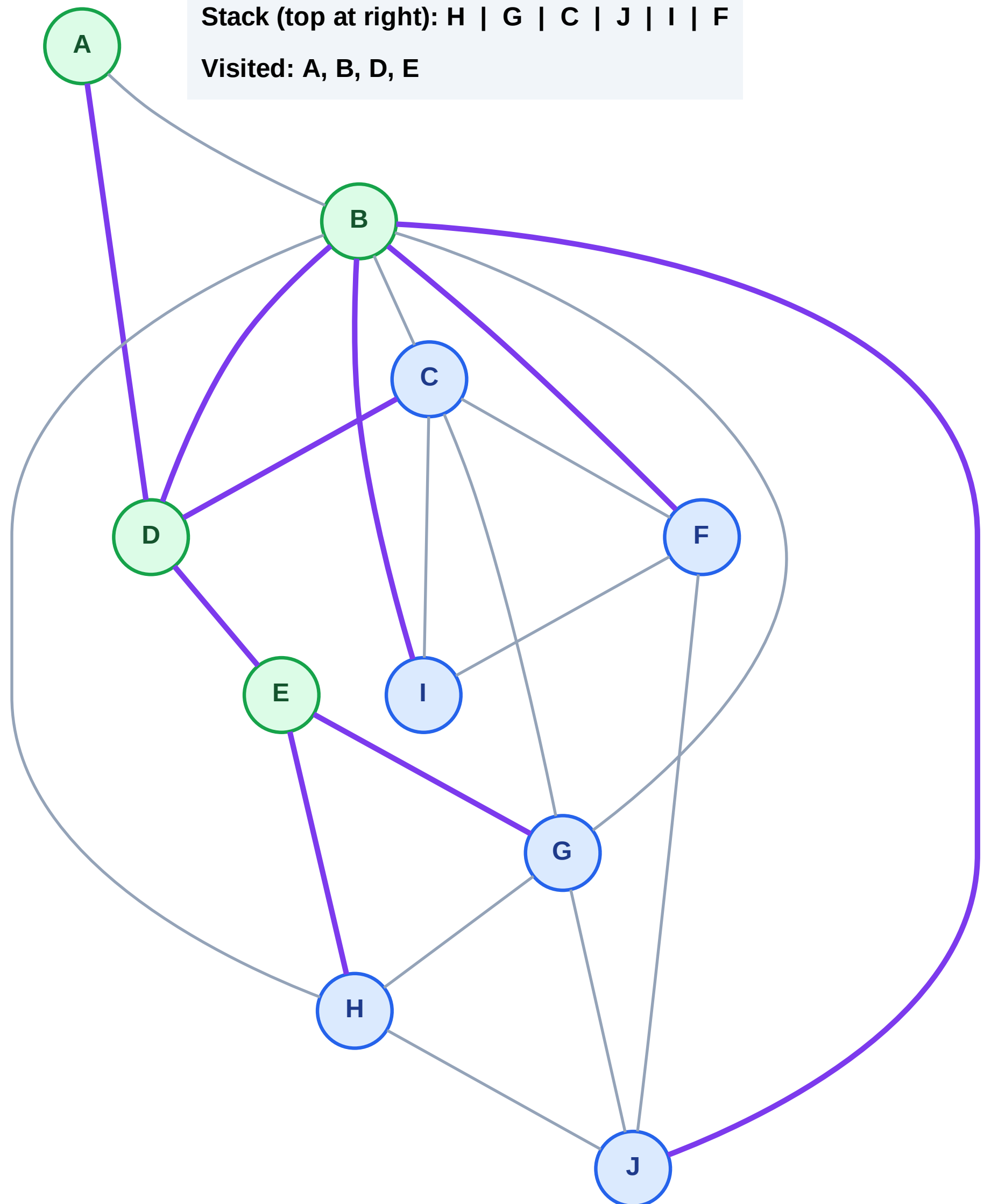
Step 9: Discover J, I, F from B

Legend

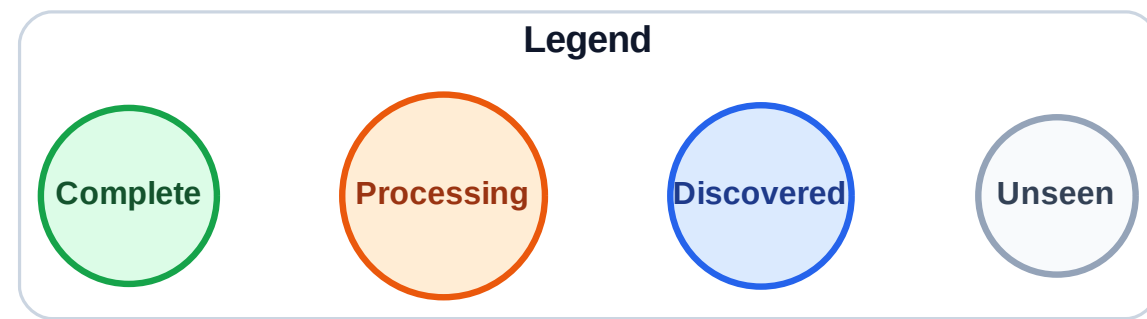


Stack (top at right): H | G | C | J | I | F

Visited: A, B, D, E

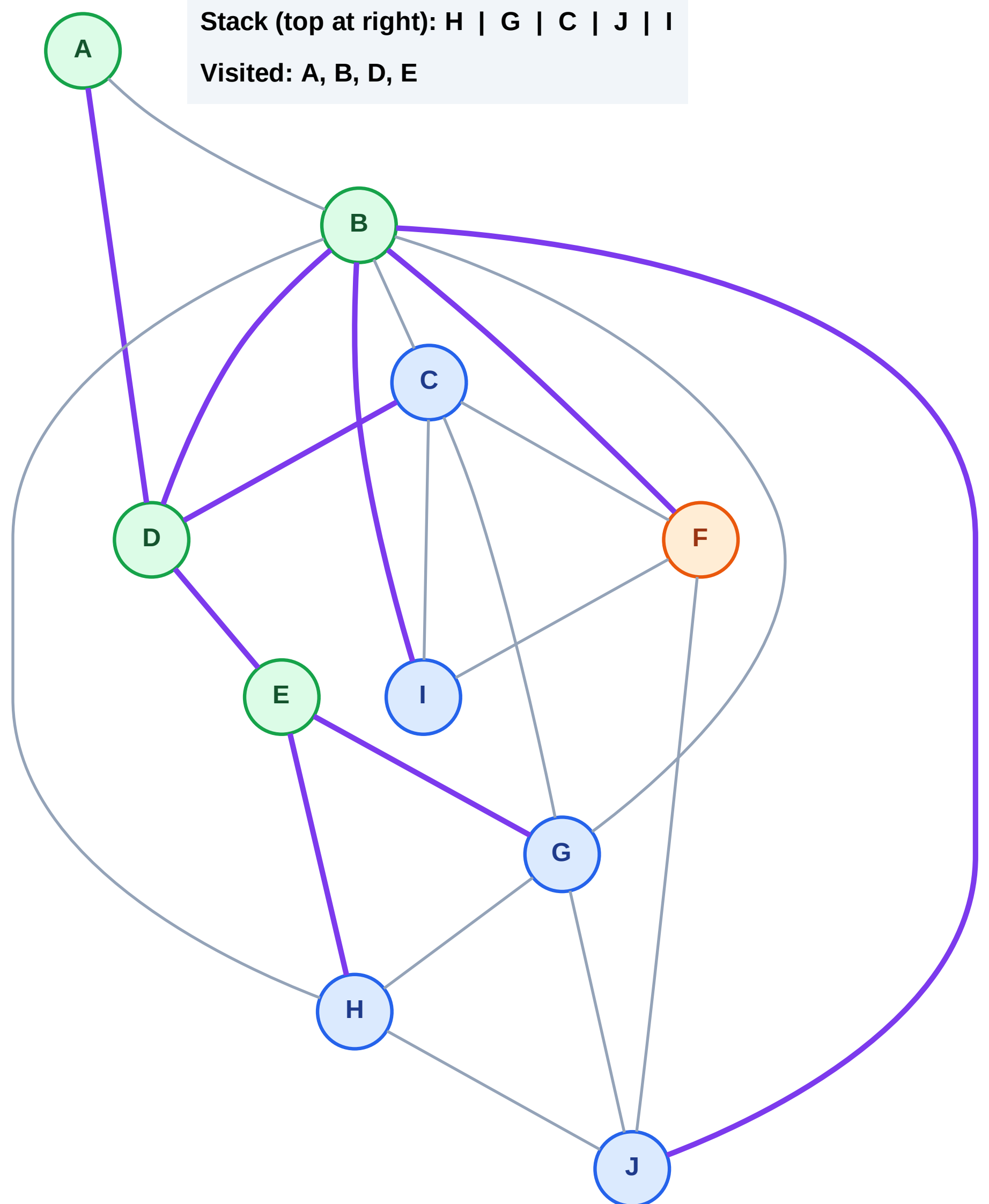


Step 10: Process node F



Stack (top at right): H | G | C | J | I

Visited: A, B, D, E



DFS Traversal

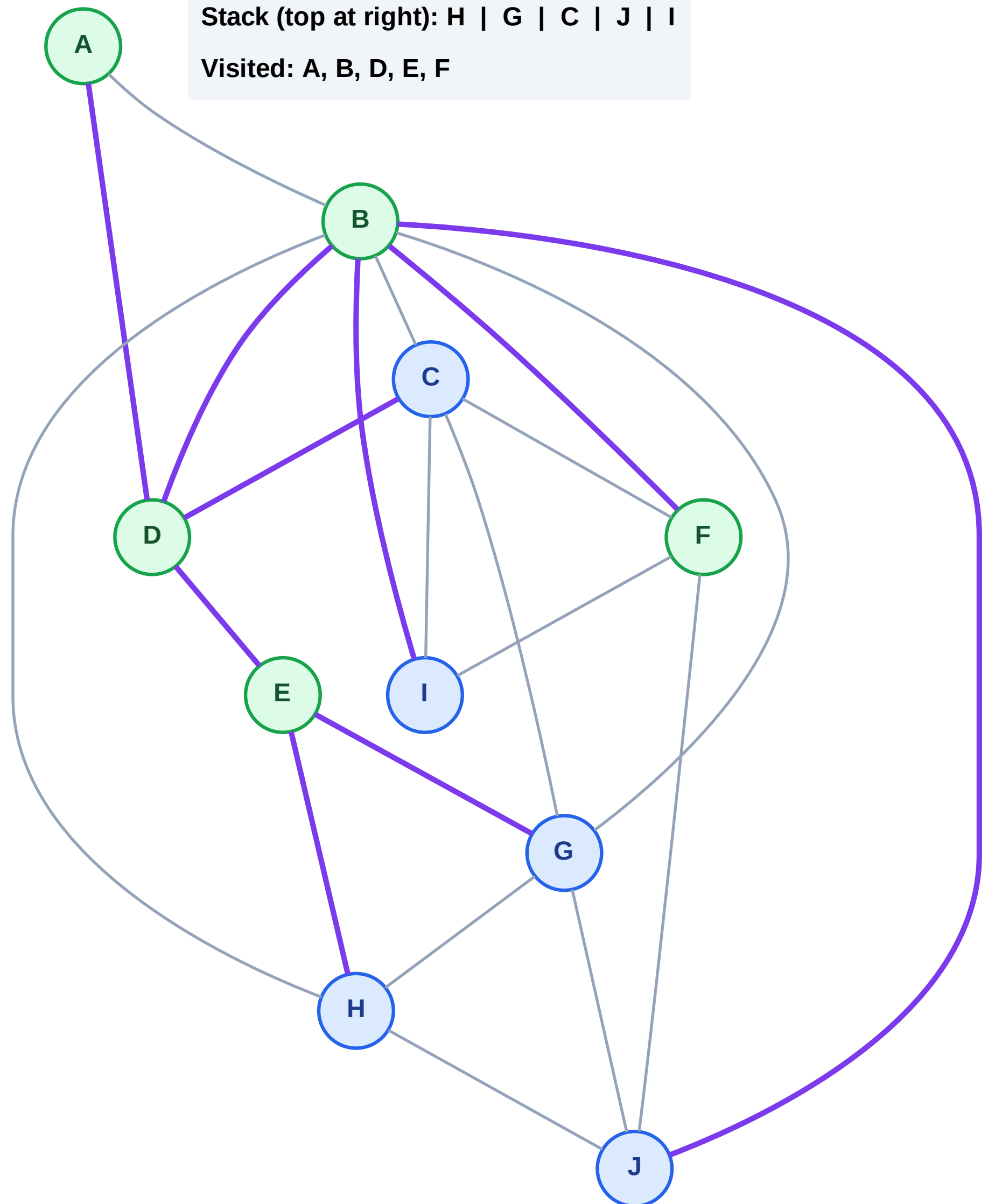
Step 11: No new neighbors from F

Legend



Stack (top at right): H | G | C | J | I

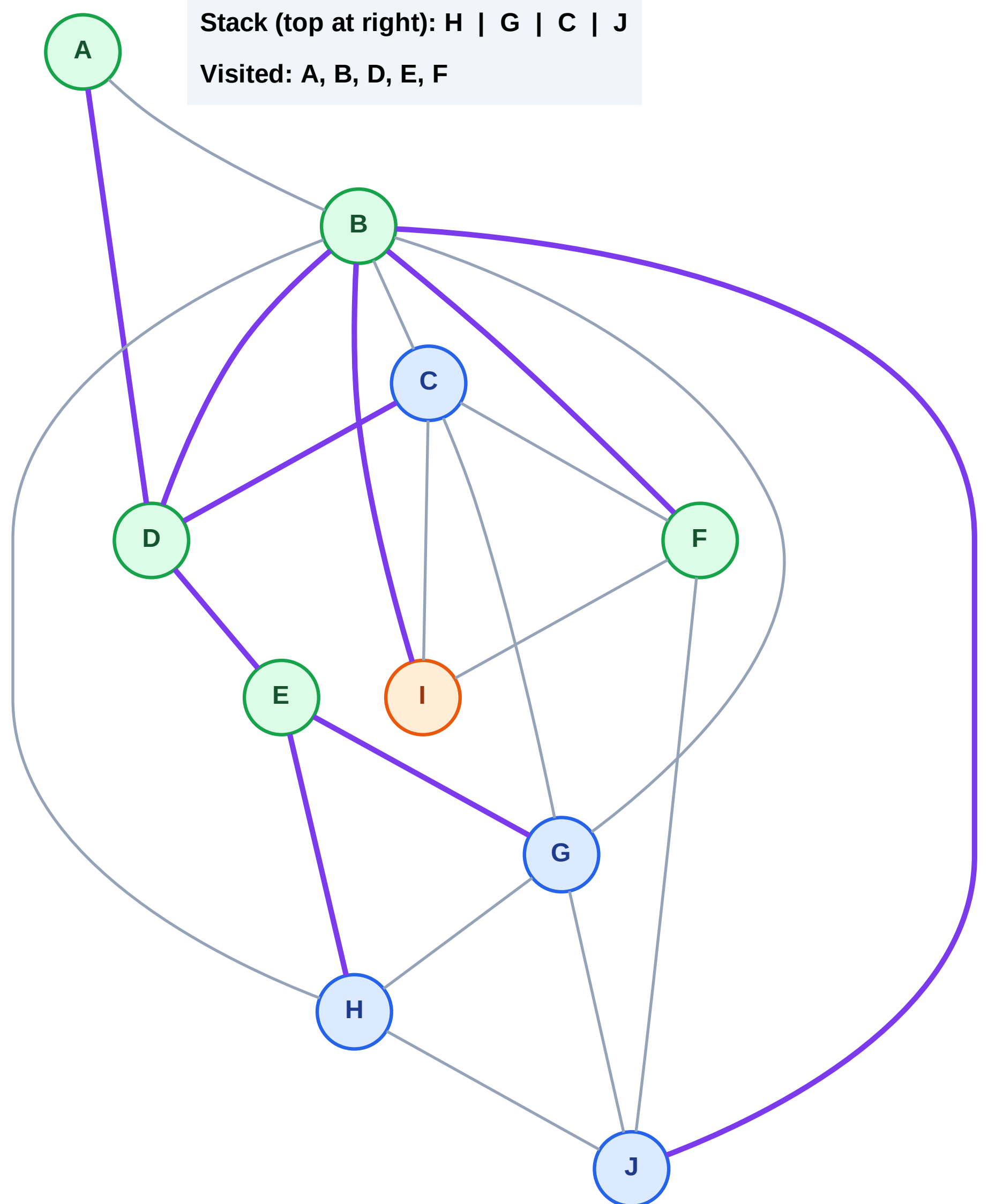
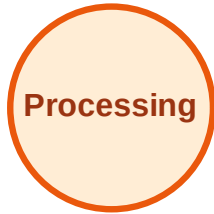
Visited: A, B, D, E, F



DFS Traversal

Step 12: Process node I

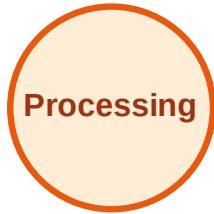
Legend



DFS Traversal

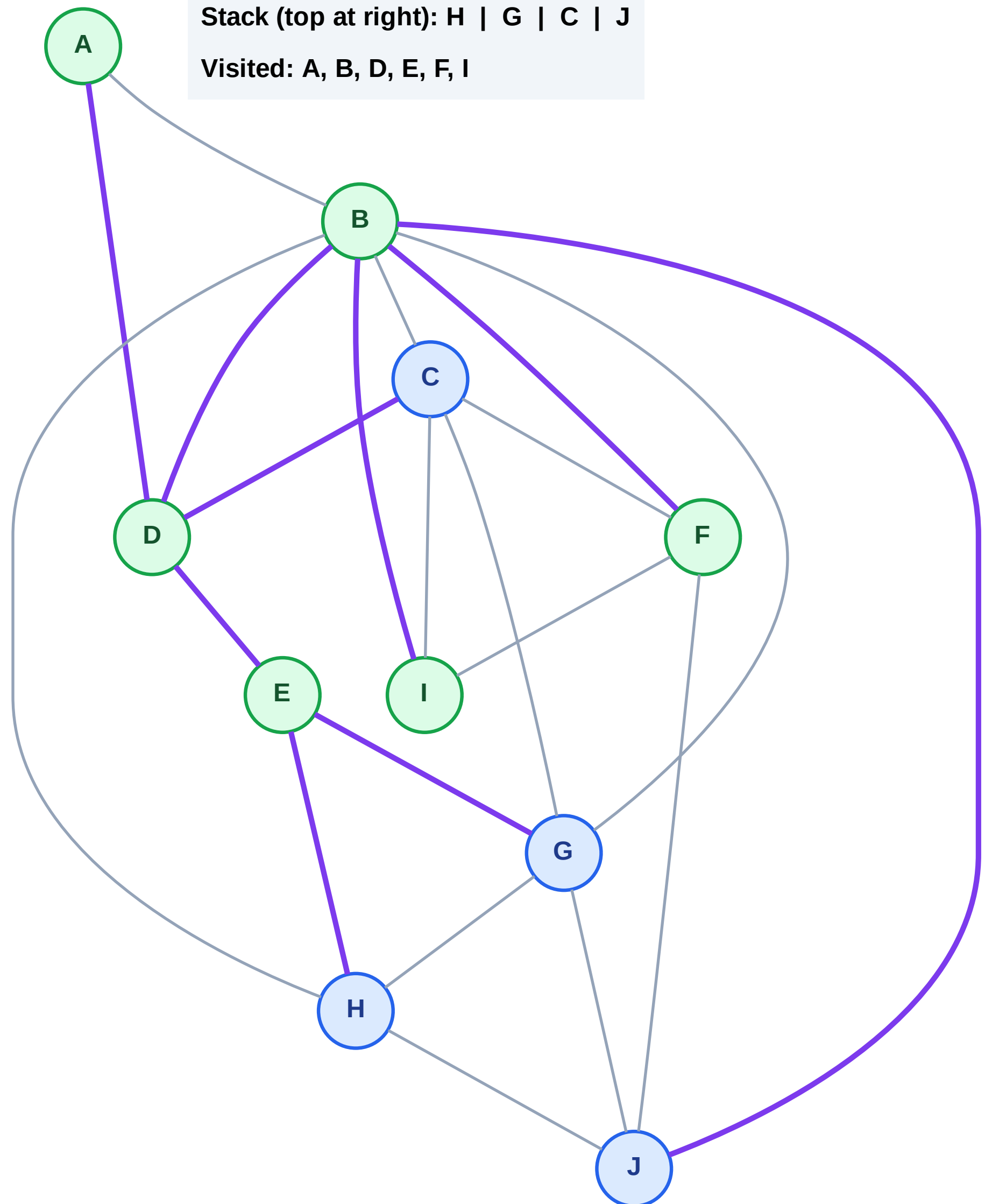
Step 13: No new neighbors from I

Legend



Stack (top at right): H | G | C | J

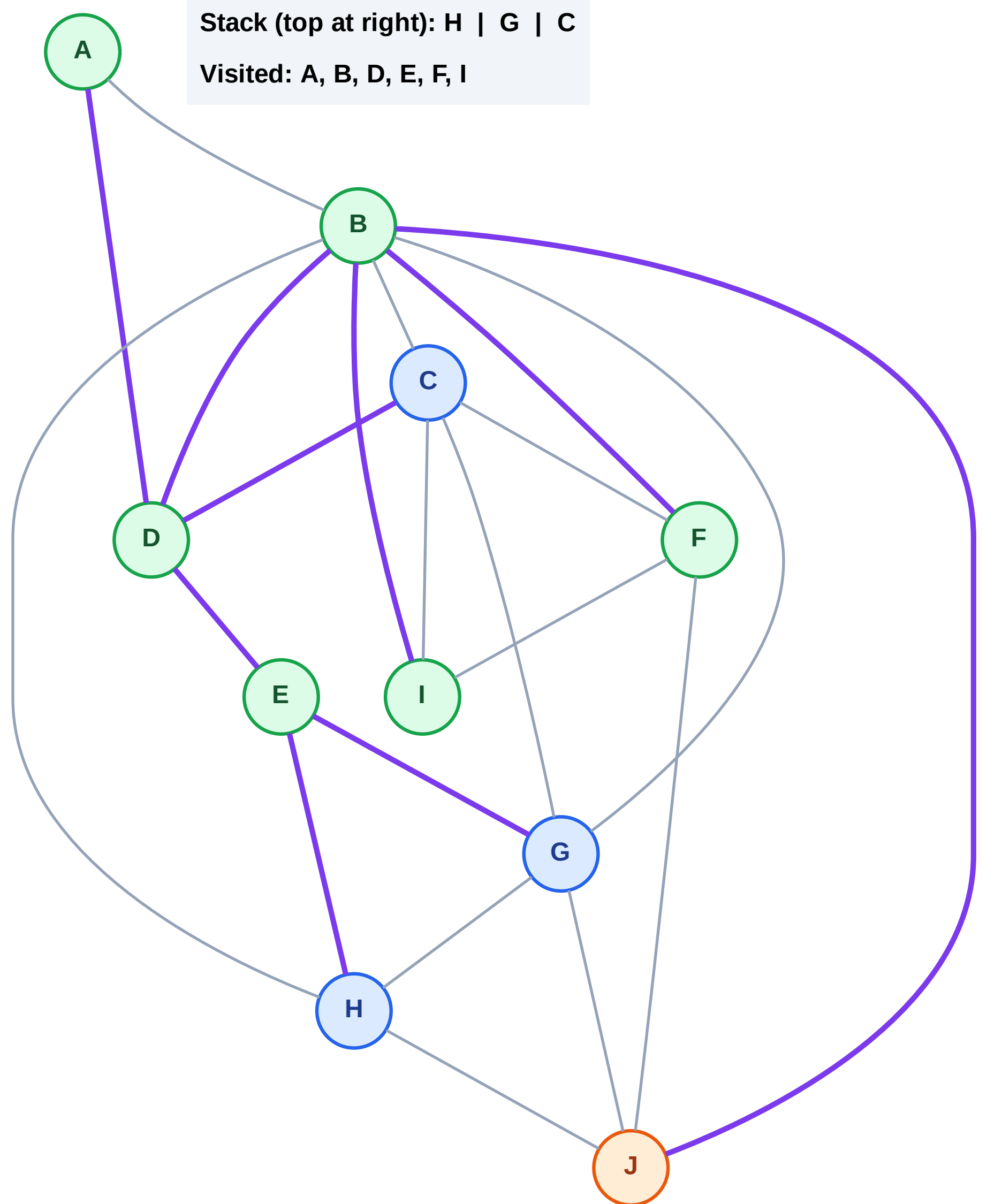
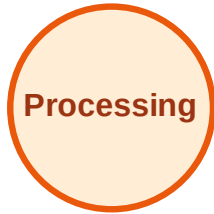
Visited: A, B, D, E, F, I



DFS Traversal

Step 14: Process node J

Legend



DFS Traversal

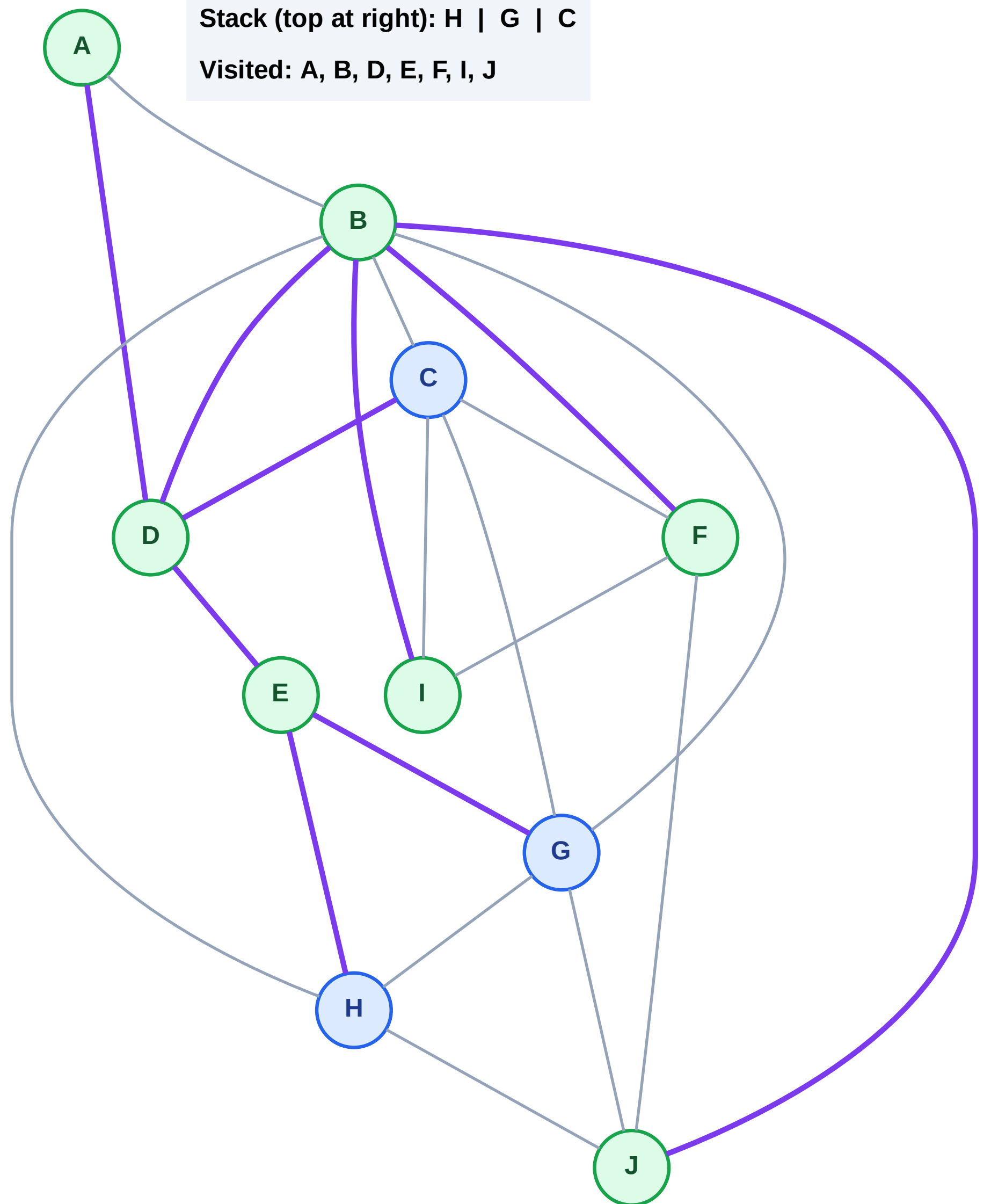
Step 15: No new neighbors from J

Legend



Stack (top at right): H | G | C

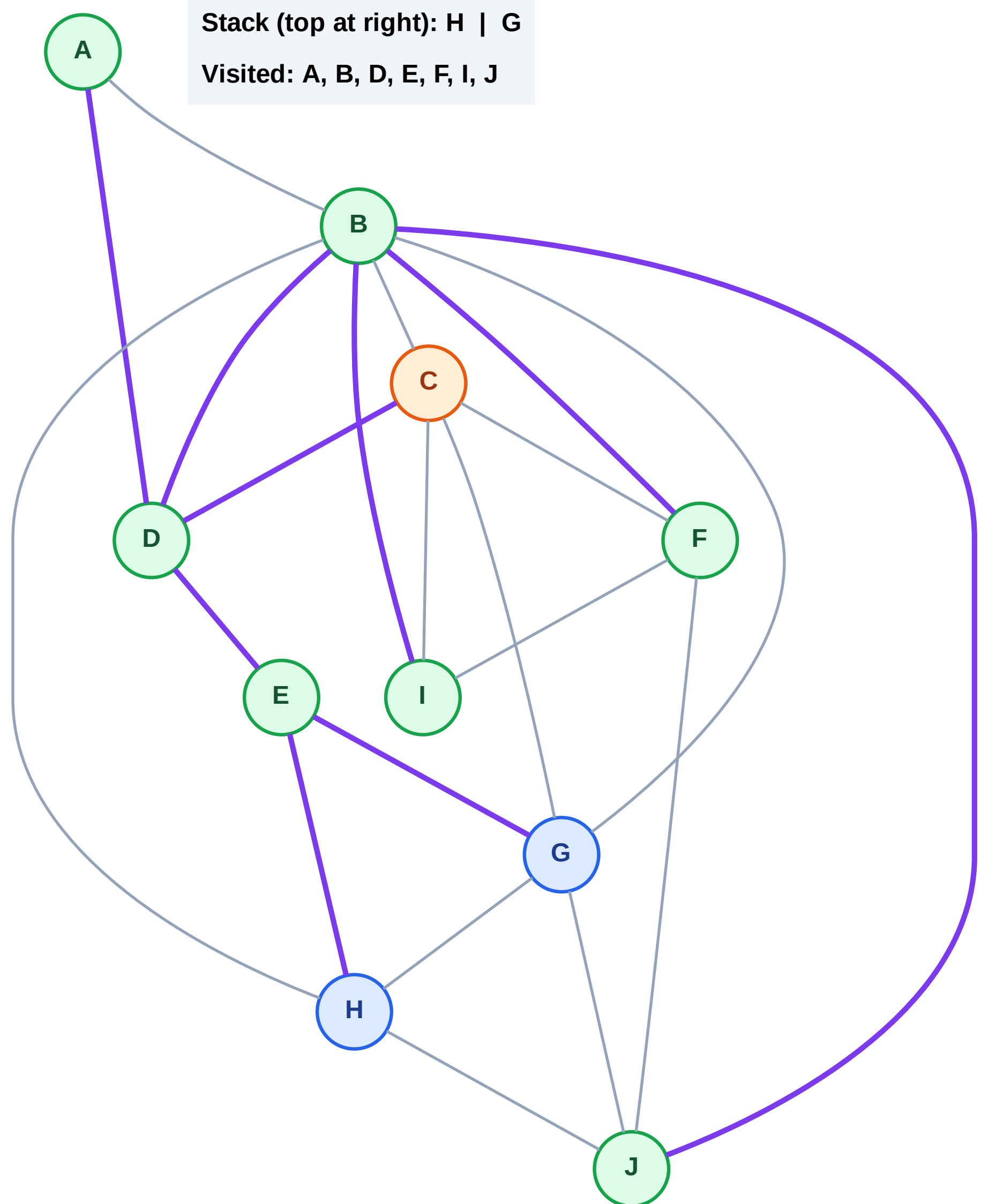
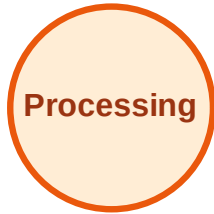
Visited: A, B, D, E, F, I, J



DFS Traversal

Step 16: Process node C

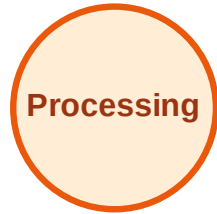
Legend



DFS Traversal

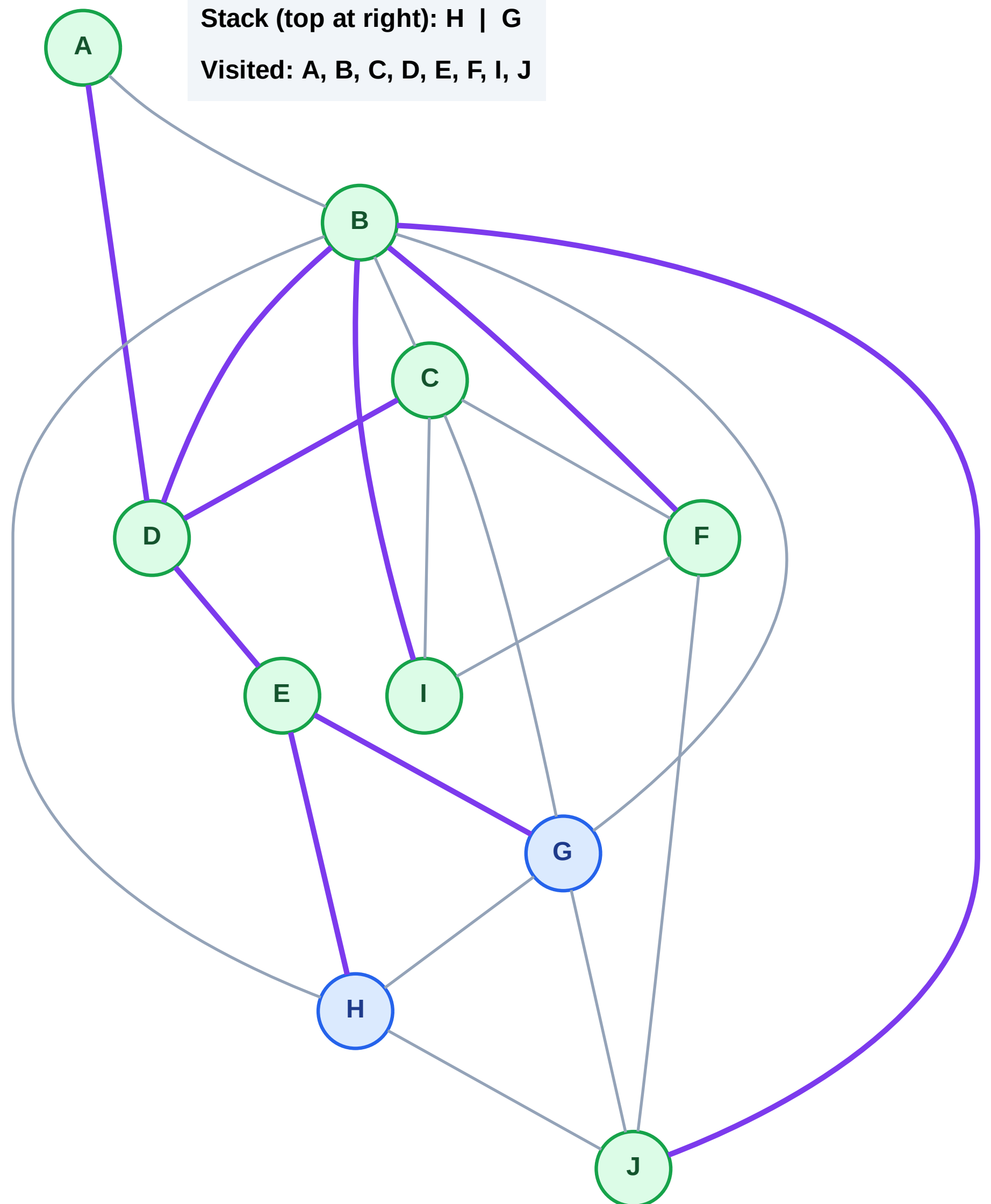
Step 17: No new neighbors from C

Legend



Stack (top at right): H | G

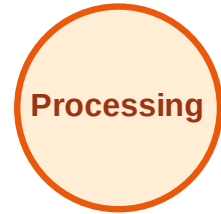
Visited: A, B, C, D, E, F, I, J



DFS Traversal

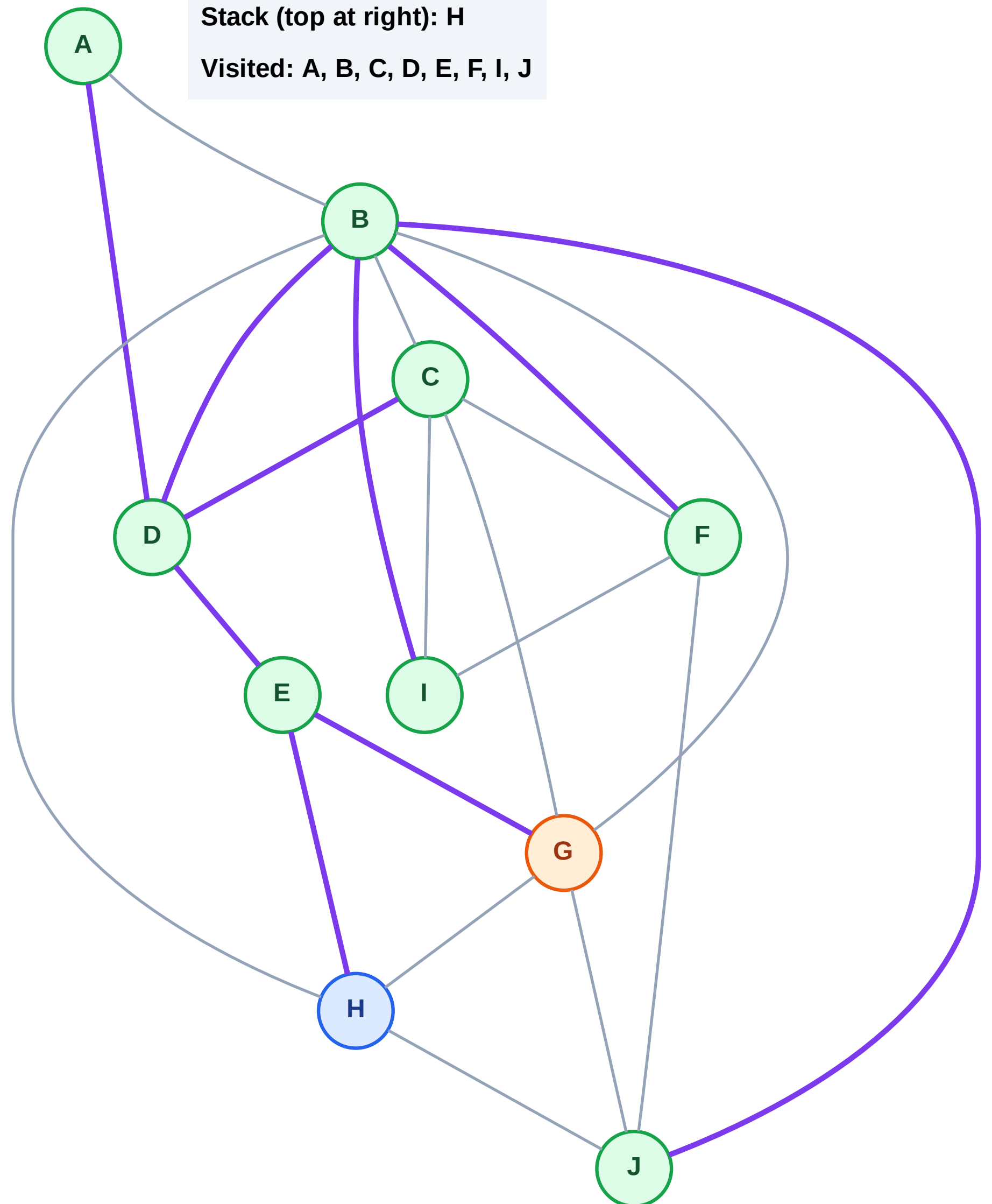
Step 18: Process node G

Legend



Stack (top at right): H

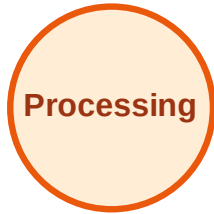
Visited: A, B, C, D, E, F, I, J



DFS Traversal

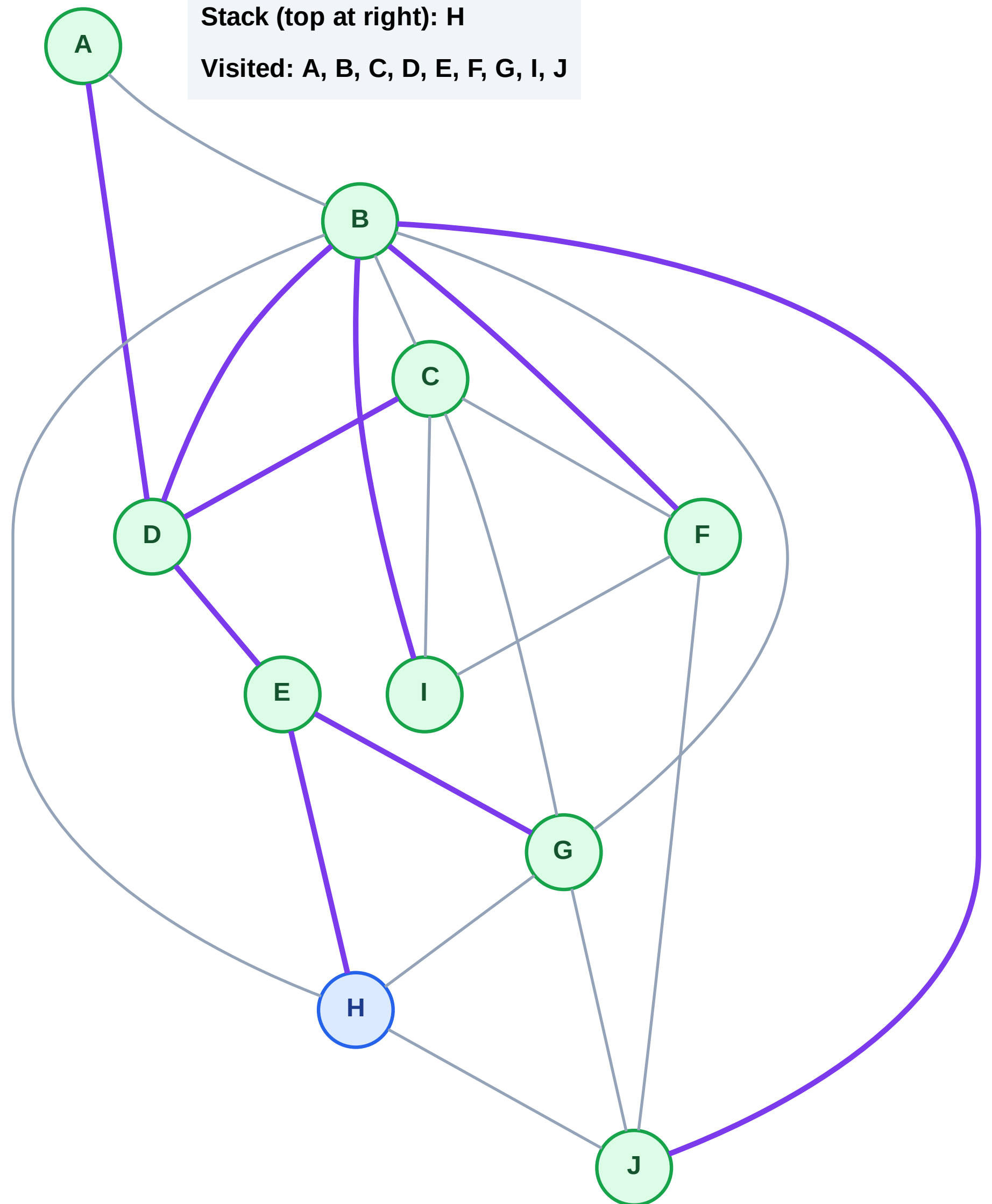
Step 19: No new neighbors from G

Legend

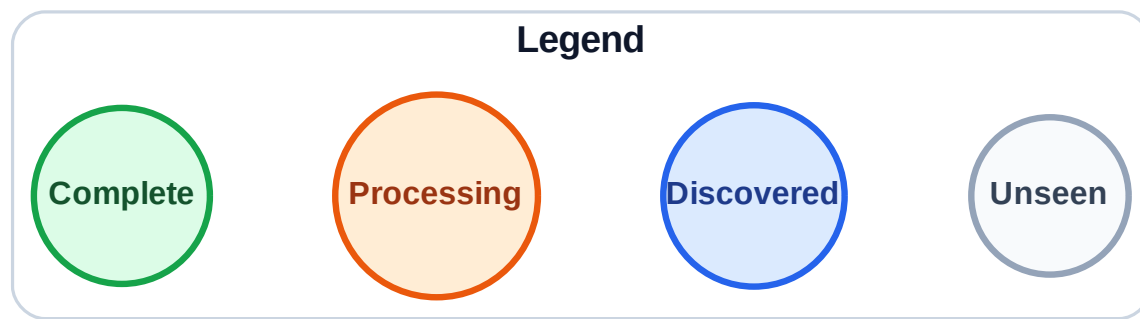


Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J

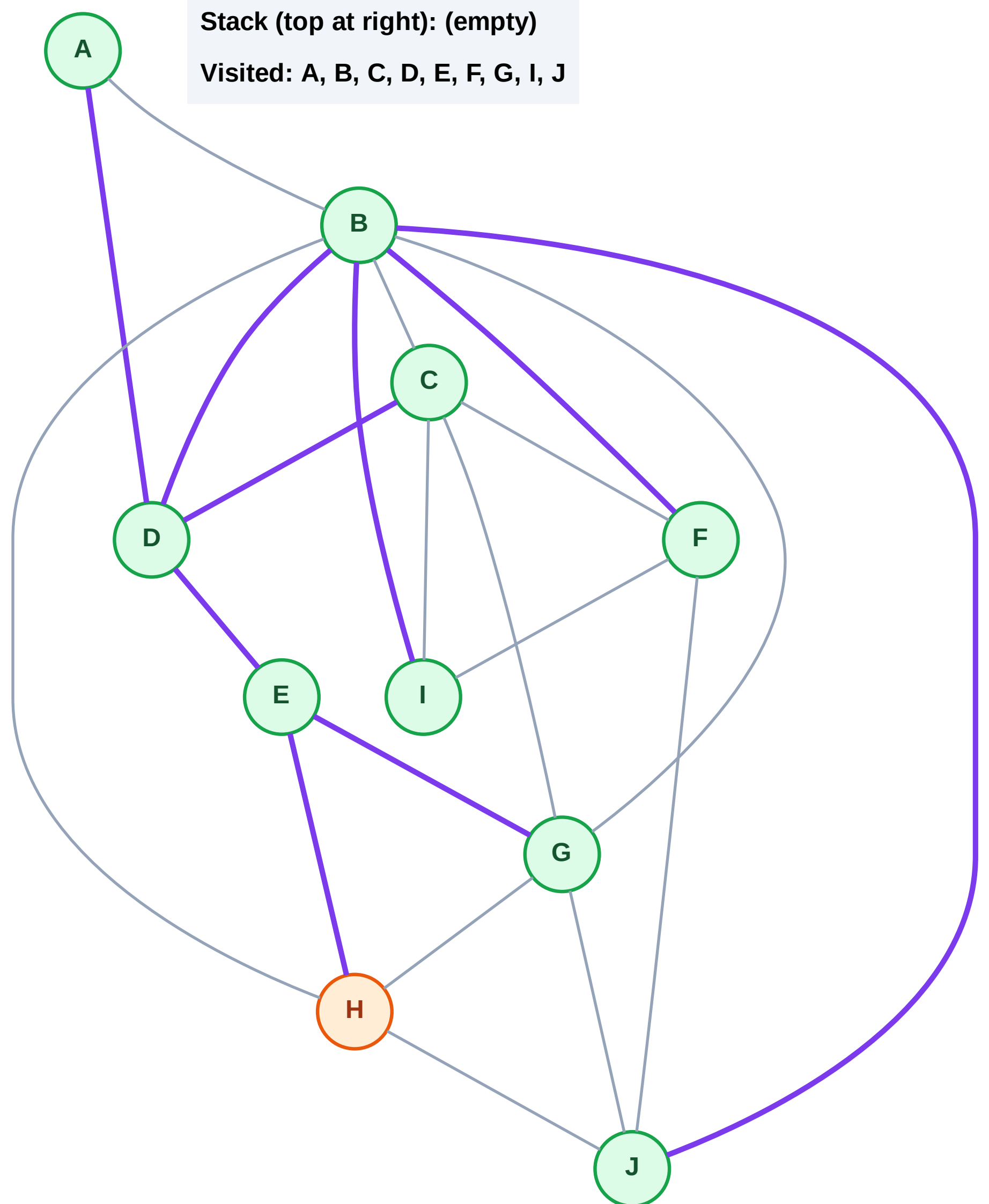


Step 20: Process node H



Stack (top at right): (empty)

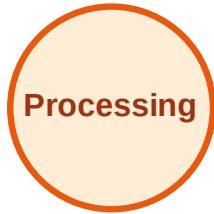
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

